

REPOSITORY-WIDE EVIDENCE AUDIT

Formal Atlas of the Research Derived Experiments System

Proofs, dynamics, empirical findings, failures, and open conjectures

115

experiment packages

129

paper sources

513

Lean declarations

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Audit and manuscript production by OpenAI Codex under direction and review

Audit date: 2026-08-19

Repository commit: 3a86730

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Executive verdict

The repository contains a real formal and empirical research system, but not one proved theorem saying that geometry explains intelligence. Its most mature mathematical center is task-relative representation: a task family induces a quotient of concrete realizations; the quotient's fibers record distinctions treated as irrelevant; a compiler chooses or generates realizations inside a fiber; and interventions, concern, viability, and access determine which realizations are reachable and useful.

Four conclusions survive the repository-wide audit.

- The finite quotient-fiber-compiler core is substantive. Lean source contains

hundreds of theorem, lemma, and proposition declarations with no actual `sorry`, `admit`, `project-local axiom`, or `native_decide`. The formal scope is often finite, discrete, algebraic, or a registered witness. It must not be silently transported to measure-theoretic, stochastic, neural, or natural domains.

- The empirical program is strongest where it performs exact finite

enumeration, declares noncompensatory gates, and preserves counterexamples. Obstruction-aware admission, relative identifiability fixtures, the SIC finite instruments, delete-repair cells, and the EML access split are the clearest examples.

- The program has advanced through failed equivalences as often as through

positive findings. Behavior is not representation; uncertainty is not current error; current error is not value of information; completeness is not access; shortest description is not always access; fiber mass is not a process-independent search law; and a catalog of repairs is not a universal calculus.

- The grand claims remain conjectural. General SIC, natural-world common

sufficient screens, universal weakness-to-OOD transport, active attractor geometry, concern as valence, real-agent selfhood or intention, open-world discovery, and general neural EML access are not established.

The audit also found status drift. The current repository policy defines verified as SafeVerify replay with headline axioms contained in `{propext}`. The latest receipt accepts `Quot . sound` for several passed headlines. Those headlines are SafeVerify-passed but do not satisfy the current literal policy. Mathlib results are Lean-proved but not SafeVerify-verified. Several older papers use verified more broadly. This atlas uses the stricter current policy and records the inconsistency rather than resolving it by fiat.

1. Epistemic types and audit method

Every claim in this atlas belongs to one of five types.

- Definition: fixes an object, map, state, score, or relation. A definition has

no truth status beyond being well-formed.

- Verified theorem: a named Lean headline with an applicable SafeVerify

receipt and an axiom footprint satisfying the repository's current literal whitelist.

- Proved, not verified: Lean elaborates with no placeholders, but the statement

lacks an accepted SafeVerify receipt, uses the mathlib lane, or has a receipt whose axiom policy conflicts with the current rule.

- Empirically supported: an exact enumeration, deterministic computation, or

preregistered stochastic result that passed its relevant gates. The claim is bounded to the registered representation, data clock, process, and controls.

- Conjecture or interpretation: an unproved generalization, philosophical

reading, causal transport claim, or proposed architectural law.

The audit treats the repository as a typed discovery graph. Artifact types are Lean declarations, papers, preregistrations, experiment manifests, committed result summaries, evidence and claim registries, SafeVerify receipts, tests, and rejected or withheld artifacts. Operations include proof elaboration, kernel replay, exact enumeration, seeded simulation, model training, intervention, adjudication, and synthesis. Gates include theorem assumptions, SafeVerify policy, preregistered fatal gates, negative controls, integrity checks, uncertainty requirements, and provenance links.

The current regime is broad but uneven. There are 115 provenance-bearing top-level experiment packages plus the shared common support package, 128 paper sources, 60 Lean files, 190 Python test modules, 67 structured experiment contracts, and 48

legacy exceptions. A package appearing in the verification index establishes traceability, not scientific truth. A result file with `status: pass` establishes that its local runner completed its recorded gates; it does not override a false scientific gate or a stricter claim registry.

2. The master formal object

Definition 2.1 - Realization system

A realization system is a tuple

$$S = (X, \text{Theta}, T, Z, q, K, C, A, G, V, E)$$

where X is a concrete realization or world space; Theta is a parameter or hypothesis family; T is a family of tasks or observables; Z is a structural state space; $q : X \rightarrow Z$ is a coarse-graining; $K : Z \rightarrow X$ is a compiler or Markov kernel; C is a concern state; A is an action set; G is a symmetry or transformation family; V is a viability region; and E is an allowed experiment family.

This tuple unifies the main programs without claiming they share one physical mechanism. Different papers instantiate only subsets of it.

Definition 2.2 - Fiber and compiler

For $z \in Z$, the fiber is $F_z = \{x \in X : q(x) = z\}$. A valid compiler is fiber-supported:

$$K(F_z \mid z) = 1.$$

The quotient says which differences are ignored at the structural level. The compiler says how a structural state is realized. Those are distinct: the same quotient can support many compiler dynamics, priors, or search procedures.

Definition 2.3 - Common sufficient screen

A quotient q is sufficient for task Y when Y factors through q . It is a common sufficient screen for task family T when every member factors through it. A coarsest common sufficient screen preserves all registered task behavior while discarding the most concrete variation.

Proposition 2.4 - Finite SIC-A construction

In the finite nonempty positive-support setting, the repository proves a construction of the master fibration. A likelihood-ratio vector against a pivot defines q ; Z is its image; a uniform-on-fiber kernel defines K ; and finite Halmos-Savage minimal sufficiency, the coarsen-refine retraction, and CS-2 give the relevant sufficiency and minimality statements. This is Lean-proved in the mathlib companion and witnessed exactly on the 4-bit Boolean world.

The result is not the general SIC-A claimed by the program's broadest prose. Standard-Borel, topological, measure-theoretic, open-support, and learned-task versions remain open.

Proposition 2.5 - Rate-distortion deformation

For registered finite sources, a distortion budget deforms the fibration from fine task-preserving structure toward a constant encoder. The uniform Hamming closed form, achievability, and converse are Lean-proved in the mathlib lane. The experimental rate-distortion pair passes its exact finite gates. This supports the finite parameterization, not a universal claim about learned neural representation paths.

Proposition 2.6 - Learnability under covering assumptions

Finite common-sufficient recovery obeys coupon-collector-style bounds under separation and fiber-balance assumptions. At resolution ϵ , an ϵ -cover reduces the continuous problem to a finite one, producing an exponential dependence on intrinsic dimension absent stronger bias. The repository's exact instruments support the bound on registered grids.

Uniform polynomial learnability in intrinsic dimension is false without additional structure. Linear ICA, sparse ICA, auxiliary-variable, and interventional settings are conditional antecedents, not a general escape.

3. Representation dynamics

Definition 3.1 - Abstraction frontier

A representation is scored by task insufficiency, dynamical non-closure, coding cost, and control regret. The non-dominated set is a Pareto antichain. This replaces the idea of a single universal ladder from concrete to abstract.

AF-1's antichain property is a direct order-theoretic fact. The registered static instrument finds a two-point frontier and confirms that identity can be dominated by a coarser sufficient quotient. The interpretation is important: more detail is not automatically a better representation.

Definition 3.2 - Obstruction

Given a candidate screen q_D and target Y , an obstruction is a pair x, x' such that $q_D(x) = q_D(x')$ but $Y(x) \neq Y(x')$. One such pair proves that Y cannot factor through q_D .

Definition 3.3 - Repair

A repair adds a coordinate, changes a quotient, or supplies transport so that the target becomes representable. The delete-repair program distinguishes:

- over-invariance: restore a distinction that was wrongly collapsed;
- under-invariance: quotient or covariantize a leftover privilege;
- covariant compensation: retain multiple charts and add transport.

Lemma 3.4 - Independent repair composition

The formal RR-2 core requires more than the informal phrase "each lift repairs its invariant." A lift must preserve an invariant once captured, and the lifts must commute. Under those assumptions, their composition ensures both invariants. Without preservation, one lift can destroy what the other fixed.

Finding 3.5 - The universal repair calculus failed

The eight-row repair table is useful but is not a universal function from a cheap failure signature to a unique minimal lift. Finite menu-blind and relabel tests kill that stronger claim. The successful `kappa_screen` uses the same common-sufficient screen as SIC plus a disclosed total order. Multiple adequate repairs can be incomparable, no largest adequate repair need exist, and greedy repair can depend on order.

The resulting house conclusion is conservative and stronger: delete-repair is motion on SIC's task-relative frontier, not a second master object.

4. Local theories, causal meaning, and gluing

Definition 4.1 - Causal meaning quotient

For a message m , context c , and downstream update $\Psi(m, c)$, define

$$m_1 \sim_{\Psi} m_2 \text{ iff for every } c, \Psi(m_1, c) = \Psi(m_2, c).$$

The quotient by this equivalence is the coarsest downstream-preserving meaning screen. A co-occurrence partition can be incomparable with it. The finite theorem is formal; the broader claim that human linguistic meaning is exactly this quotient is an interpretation.

Definition 4.2 - Theory atlas

Local contexts carry charts M_i with transition maps T_{ij} . A cocycle discrepancy compares direct and composed transitions on triple overlaps. In the registered finite classifier, discrepancy support separates three cases: glue, a localized boundary, and a distributed missing-latent signature.

Formal boundary 4.3

The SafeVerify-verified object is the finite TA-2 classifier and registered witnesses. The naked paper-level TA-1 "cocycle iff gluing" needs injectivity or an equivalent condition; a constant family is a counterexample to the unqualified version. The smallest alphabet enlargement that repairs an arbitrary failed cocycle is unproved.

5. Concern, compiler ecology, and viability

Definition 5.1 - Concern reweighting

Given utility or concern statistic U_c , concern changes a compiler inside an existing fiber:

$$K_c(x \mid z) \text{ proportional to } \exp(\beta * U_c(x, z)) K(x \mid z).$$

The support remains inside the fiber. Concern therefore selects among realizations without by itself changing what the quotient represents.

Proposition 5.2 - Finite concern geometry

For the registered finite exponential-family model, the Fisher information is the covariance of the sufficient statistic, scaled by β^2 . The repository also proves a finite rectangular holonomy/area identity. These are mathlib proved-not-verified results. Their interpretation as contextual meaning, valence, or lived concern is not proved.

Definition 5.3 - Compiler ecology

A compiler population can evolve by a Boltzmann update:

$$K_{t+1}(x \mid z) \propto K_t(x \mid z) \exp(\beta \cdot r_t(x, z)).$$

The finite Chebyshev/Boltzmann core proves that nonnegative reward tilt does not decrease expected reward under the stated assumptions. A Bayesian compiler update coincides with this form when reward is log likelihood. Full stochastic MDL consistency and broad evolutionary interpretations remain open.

Definition 5.4 - Viability dynamics

Let $V \subseteq Z$ be viable. A bounded agent attempts to keep $q(X_t)$ in V while learning which effects are self-caused, world-caused, or jointly caused. The most defensible maintained-concern loop is

```
disturbance -> detect -> allocate intervention -> update attribution
               -> regulate -> cool or satiate -> quiesce -> re-engage
```

This is a state machine assembled from synthetic experiments. It is not a theorem of consciousness, general agency, or selfhood.

6. Weakness, symmetry, and OOD generalization

Definition 6.1 - Symmetry-compatible weakness

$W_G(f)$ counts or weights transformations in G for which a hypothesis transports consistently with an allowed output action. Large compatible volume is "weak" because the hypothesis commits to fewer arbitrary distinctions while remaining valid.

Empirical finding 6.2

On registered shortcut-compatible symbolic and neural families, weakness is a stronger predictor of OOD accuracy than the implemented loss, description length, compression, sharpness, norm, and validation baselines. The largest registered neural sweep contains 4,096 models; true-group weakness has Pearson correlation about $+0.8085$, compared with $+0.2533$ for parameter norm and $+0.0924$ for validation. Structure-compatible semantic selection reaches about 0.978 OOD versus 0.919 for random or ID selection and 0.751 for a wrong structure.

Failure 6.3 - External portability

The external Pythia LoRA design hard-rejects portable weakness-to-generalization causality in that setting: mean OOD is about 0.0285 , weakness correlation is about -0.0817 , and a classical comparator is stronger. The internal result therefore remains real but domain-bounded.

Formal boundary 6.4

A two-point finite coverage proposition and mixture-prior mass facts are SafeVerify-replayed. The general group-completion statement, the Langford-Seeger-Maurer PAC-Bayes inequality, neural parameter-space transport, and OOD causality are not proved by those artifacts. The PAC-Bayes experiment's top-level pass coexists with one false OOD wrong-group gate; its own claim boundary withholds OOD transport. This atlas follows the gate, not the word pass.

7. Access is process-relative

Definition 7.1 - Expressivity and access

For a syntax language, q maps terms to denotations. A fiber contains all terms with the same denotation. Expressivity asks whether the fiber is inhabited. Access asks whether a specified process can reach a useful member. Formula size, circuit size, prior mass, optimization basin, and rewrite basin are different objects.

Proposition 7.2 - Four-seam separation

The Mul/Sq monomial toy separates four seams exactly: both languages express the target; tree size differs exponentially; shared circuit size does not; and normalized Gibbs fiber mass differs sharply. At $n=4$, the registered $\log_2$ mass ratio exceeds 28. The Lean core proves the exact tree and circuit facts; the finite instrument checks Catalan shell counts and normalization.

Experiment sequence 7.3 - EML US-4 prime

- Truncated Gibbs mass distinguishes two same-minimum-size targets by 2.015625, so shortest depth is insufficient.
- Matching-skeleton gradient descent succeeds 8/8 versus 6/8, a bounded local result consistent with the mass ranking.
- Unknown-skeleton gradient descent ties 7 versus 7; mass does not transfer to this freely retunable search process.
- Frozen-leaf greedy rewrite produces 43 versus 28 extra basins; the effect returns when the process preserves discrete structure.

The conclusion is not "fiber mass governs search." It is that access is a property of a fiber plus a process. The neural bootstrap and any general law mapping Gibbs mass to gradient recovery remain open.

8. Identification, discovery, and admission

Definition 8.1 - Relative identifiability

Allowed experiments E induce observational equivalence on hidden worlds. Target τ is identifiable exactly when it factors through the observational quotient. A target-disagreeing indistinguishable pair is a complete obstruction certificate. Adding experiments can only refine the quotient.

The Lean package proves the quotient/factorization kernel for arbitrary dependent experiment systems. It does not prove statistical identifiability under noise, misspecification, continuous spaces, or limited samples.

Definition 8.2 - Typed scientific outcome

A bounded scientific controller should return one of four types:

- recovered target;
- terminal obstruction;
- recoverable but outside the current budget;
- a next experiment on an optimal worst-case branch.

This prevents a lucky guess, a resource failure, and a mathematical impossibility from sharing one score.

Empirical finding 8.3 - Exact obstruction-aware admission

The registered exhaustive screen evaluates 500,912 binary systems and 1,975,104 hidden-world episodes with zero mathematical, certificate, recovery, or oracle-dominance failures. It preserves 26,304 strict counterexamples to immediate target-pair greedy choice; the smallest registered case costs 3 under greedy choice versus an optimal worst-case cost of 2.

This is exact finite decision control. It is not new optimal-decision-tree mathematics, a scalable open-world planner, natural scientific discovery, or universal agency.

9. The empirical correction chain

The long concern/agency arc is best understood as a sequence of corrected identifications.

- Behavior is not representation. A controller can act correctly while the proposed hidden decomposition is wrong or unmeasured.
- Representation is not competence. Decodable geometry can exist without the

intervention-sensitive mechanism needed by the task.

- Uncertainty is not error. Bootstrap ensembles can agree at a regime boundary

where all members are confidently wrong.

- Residual magnitude is not systematic error. Irreducible Bernoulli noise can

dominate a probe-value signal.

- A historical residual is not current error. Debiasing can remove noise yet

preserve stale or unevenly sampled calibration.

- Current error is not value of information. Exact Bayesian enumeration gives

counterexamples where the most informative misspecified probe has zero oracle EVSI and negative true regret reduction.

- Re-engagement is not stable re-engagement. A trigger can fire once yet fail

latency, specificity, or repeated-disturbance gates.

- Total prediction is not component identifiability. Self/world heads can be

gauge-symmetric while total outcome prediction stays accurate.

This correction chain is one of the repository's strongest findings. Each failed equivalence narrows the object that a later experiment must identify.

10. Supported empirical findings by program

10.1 Finite formal instruments

The abstraction frontier, causal semantics, common sufficiency, rate-distortion, finite SIC-A witness, representation repair, theory atlas, compiler ecology, delete-repair suite, concern selection, silent substitution, and related exact packages mostly pass their registered finite gates. Their value is executable scope discipline: they demonstrate consistency and expose missing assumptions. They are not independent evidence for natural-world SIC.

10.2 Concerned syntax and viable bodies

Concerned syntax passes its registered multi-seed symbolic/vector gate when intervention use is gated by task concern. Viability-guided typed bodies pass the joint formal, resource, action, and anti-cheat surface while shortcut controls fail. These are bounded synthetic acceptance surfaces, not solved natural perception, open semantics, or neural architecture search.

10.3 Causally grounded finite agents

World-responds Suite C passes its 64-seed bootstrap gate with recovery 1.0 and a large selectivity lift. Long-horizon/tool experiments establish several behavioral and memory surfaces, but also preserve provider-specific repair failures and hidden-site negatives. The benchmark charter's strongest honest claim is methodological: final-answer accuracy should be paired with at least one structure-specific gate and anti-cheat control.

10.4 Hysteresis without bifurcation

The preregistered passive-active phase map rejects a reproducible discontinuous bifurcation: critical-point stability and order-parameter co-location fail. A separate matched-budget continuation result supports path dependence: loop area is about 0.04471, six contiguous couplings are significant, washout preserves four, and reinitialization clears the effect. The correct statement is synthetic hysteresis without established bifurcation or attractor basin.

10.5 Concern-gated retrieval

The E1 synthetic pilot passes 192 deterministic episodes: coincidence hit@1 is 1.000 versus a best one-sided rate near 0.0052, verifier precision and recall are 1.000, and the maximum PPR residual is tiny. Yet additive scoring ties the product in two regimes and learned concern is already at ceiling. Multiplicative necessity, learned care, selfhood, and live-agent transfer are withheld. Later E2 waves expose ordering leakage, a killed wave, and only a partial go decision.

11. Major failures, nulls, and invalid runs

11.1 Weakness external contact

The external Pythia LoRA transfer rejects the portability claim. Internal symmetry-compatible success cannot be promoted to an external causal law.

11.2 Constraint-swap causal geometry

All five causal-geometry gates fail even though competence is 1.0 . Correct answers do not imply that the proposed representation carries the causal mechanism.

11.3 Commitment-surface generator generalization

E5 favors labeled coverage rather than generator structure: one coverage-local OOD score is about 0.741 while the generator score is about 0.063 . Generator, group-specificity, and transport gates fail. E6 is blocked at its smoke gate. E7 has no scientific verdict because 6 of 32 matched groups violate the frozen runtime budget.

11.4 Activation geometry

Small Pythia-70M pockets leak target-family controls and fail replication at Pythia-160M. A target direction can move intended relations while also moving random relations. The evidence is model-specific and nonspecific, not a causal activation-space geometry law.

11.5 Load-bearing prose

The load-bearing score is high, but the preregistered paraphrase invariance gate is killed. The result is a bounded null: prose can contain commitment surfaces on the tested substrate without surviving the registered paraphrase gauge.

11.6 Seed promotion

The deterministic calibration package rejects three-seed promotion. Three seeds are insufficient in every registered regime; independent regimes require about 16, hierarchical regimes about 64, and the weak high-noise regime remains unpromotable. A local pass means the counterexample instrument worked, not that three seeds are scientifically adequate.

11.7 Value-of-information surrogates

Current error, posterior variance, error-squared heuristics, and mutual information do not universally rank probes by true expected regret reduction. The exact enumeration passes because it preserves the counterexample. The surrogate-generalizability claim is rejected.

11.8 Neural group discovery procedure sensitivity

Data-inferred group weakness retains much of the oracle signal in the registered rotation setting, but the result depends on selection procedure. Threshold selection can tie; top-k selection changes encoder-versus-pixel rankings. A dense random-rotation control is not a strict null because it inherits partial alignment by construction.

12. Formal theorem families and claim boundaries

The formal source is organized into three lanes.

12.1 Relative-identifiability lane

This lane proves equivalence-relation laws, factorization iff fiber constancy, obstruction iff failure to factor, refinement under richer experiments, and empty or constant edge cases. Its truth is quotient theory for deterministic dependent experiment systems. Noise and statistical rates are outside scope.

12.2 Lean core lane

The core lane includes common sufficient screens, finite counting, refinement, union-bound and pigeonhole facts, causal semantics, abstraction frontiers, alignment arithmetic, theory-atlas classifiers, representation repair, delete-repair witnesses, EML enumerations, concern choices, silent substitution, and finite identifiability witnesses. Some headline subsets have SafeVerify receipts; many supporting declarations only have a green Lean build. File presence is not itself a receipt for every declaration.

12.3 Mathlib lane

The mathlib lane proves real exponential and logarithmic bounds, Chebyshev and Boltzmann inequalities, finite positive-support Halmos-Savage, uniform-Hamming rate-distortion including Fano/Jensen converse, coarsen-refine identities, finite exponential-family Fisher identities, rectangular holonomy, Bayesian mixture bounds, finite SIC-A, covering calculations, and a weakness KL certificate. It uses standard axioms such as `Classical.choice` and is explicitly proved-not-SafeVerify-verified.

The PAC-Bayes KL inequality is cited and accepted as a hypothesis in the plug-in theorem; it is not proved locally. The general ICA theorem, probability-space conditional independence, full CT-1 MDL rate, and several paper-level analytic transports remain outside the current Lean statements.

13. System dynamics as a research process

The repository itself is an adaptive system. Let a research regime be

$$R_t = (A_t, O_t, G_t, S_t)$$

where A_t is the set of representable artifact types, O_t the allowed operations, G_t the gates and verifiers, and S_t the evidence store.

A run produces candidate artifacts and residuals:

$$(\text{accepted}, \text{rejected}, \text{residual}) = \text{Gate}(R_t, \text{Run}(R_t, \text{hypothesis})).$$

Retrieval adds an already representable artifact. Search explores within the same schema. Discovery requires a stable change to an accepted artifact type, operation, verifier, grammar, mechanism, or explanatory residual. A failed candidate can therefore be productive: it changes the obstruction map even when it does not expand the accepted theory.

The program's characteristic transition is:

```
equivalence claim
-> preregister a discriminator
-> find a counterexample or boundary
-> split the object into typed components
-> build a narrower instrument
-> preserve both the supported subclaim and the rejected transport
```

This pattern explains the progression from representation to competence, uncertainty to error, error to value, completeness to access, and repair table to task-relative frontier.

14. Hypotheses and what they have borne out

H1 - Task-relevant structure is quotient-like

Status: strongly supported in finite formal and synthetic settings. The coarsest common sufficient screen, causal meaning quotient, relative identifiability quotient, and finite SIC-A all instantiate the pattern. Natural task ensembles and learned open-world screens remain unproved.

H2 - Geometry is the portable language of constraints

Status: useful organizing conjecture, partially supported by synthetic metric, fiber, chart, and intervention constructions. It is not a theorem and has failed to transfer cleanly in activation-space and external-model settings.

H3 - Weakness predicts transport better than generic simplicity

Status: strongly supported inside registered symmetry-compatible shortcut families; rejected as a current external portability law. The formal coverage kernel is much narrower than the empirical hypothesis.

H4 - Passive geometry becomes an active attractor under coupling

Status: not established. Passive and intervention-sensitive geometry exist in bounded settings; the discontinuous bifurcation claim fails. Synthetic hysteresis survives, but basin and return dynamics remain open.

H5 - Concern allocates representational or compiler capacity

Status: exact finite reweighting, Fisher, boundary, and selection effects are supported. The bridge to learned real-world care, valence, or phenomenology is conjectural.

H6 - Two-sided context and care are multiplicatively necessary for recall

Status: the coincidence mechanism beats one-sided policies in the E1 synthetic world, but additive scoring ties it in two regimes. Multiplicative necessity and learned care are withheld; later leakage failures require a more sealed design.

H7 - A universal repair calculus maps failures to minimal lifts

Status: rejected in the strong form. The finite repair table works on its rows, but menu blindness, relabeling, nonuniqueness, and order dependence kill a cheap universal function. The surviving theory is task-relative frontier movement.

H8 - Description complexity controls access

Status: rejected as a general statement. Tree size, circuit size, Gibbs mass, and process-specific basin geometry separate. Access is process-relative.

H9 - Bounded agents should detect mathematical obstruction before acting

Status: exactly supported in finite deterministic worlds. Recovery, obstruction, budget infeasibility, and optimal next action can be separated. Open, stochastic, misspecified, and large natural systems remain future work.

H10 - Maintained concern requires self/world attribution and selective

intervention

Status: a bounded synthetic mechanism exists, but architecture and calibration failures remain. The program supports computational precursors, not a claim of selfhood or consciousness.

H11 - Specifications can stay green while objectives silently drift

Status: proved for the registered zero-leakage finite kernel. Real specifications' leakage and real delegate behavior remain open empirical halves.

H12 - Discovery is a regime transition rather than high-scoring search

Status: a methodological rule supported by the repository's own correction history. It is not yet an externally validated detector of scientific discovery.

15. Documentation and status contradictions

The audit preserves the following contradictions as findings.

- Current policy says SafeVerify plus axioms contained in `{propext}` is needed

for "verified." The Wave 5-12 receipt accepts Quot . sound and labels several such headlines passed. Those rows require policy clarification.

- Several papers say all named theorems are machine-checked or Lean-verified,

while the current theorem backlog still marks general SIC-A, probability-space T4, classical T7, CT-1 MDL, naked TA-1, and other claims prose-only.

- The structural-intelligence-foundations paper calls mathlib predecessors

verified; the latest receipts classify the mathlib lane as proved-not-verified.

- The Theory Atlas paper's unqualified gluing iff is superseded by an honesty

ledger requiring injectivity or an equivalent guard.

- The top-level TODO contains stale tasks to install Lea and run initial proof

waves even though later receipts record a local environment and many replayed waves.

- Early EML prose says US-4 prime is untested; later experiments establish a

process split. Only the intended neural bootstrap and general law remain open.

- PAC-Bayes and some counterexample instruments use `status: pass` while the

promoted scientific claim is false or withheld. Runner success and scientific adjudication must remain separate fields.

- Legacy paper-only packages are indexed for provenance but often lack a structured manifest, exact result summary, or independent adjudication. They should not be counted as equal-strength evidence.

16. Open conjectures and decisive next tests

Formal residuals

- General measure-theoretic SIC-A and probability-space T4.
- A fully scoped classical ICA theorem rather than a finite signed-permutation class witness.
- CT-1 probabilistic MDL consistency and its quantitative rate.
- TA-1 with exact necessary assumptions and a minimal cocycle-closing enlargement theorem.
- Continuous or unique repair calculus statements with interacting-lift pushouts.
- A locally proved PAC-Bayes KL theorem or an explicit imported theorem object.
- A process-indexed access theorem that states when fiber statistics transport to a sampler, optimizer, or rewrite system.

Empirical residuals

- A sealed learned-geometry concern-gated retrieval experiment without role or order leakage, followed by natural transfer only if the synthetic gates pass.
- Causal activation-space interventions that beat random-relation and wrong-subspace controls across model scales.
- A direct attractor test measuring basin volume, return dynamics, and perturbation recovery rather than proxy discontinuities.
- Open, stochastic, misspecified adapters for relative identifiability and obstruction-aware admission.
- Natural images and open semantics for concerned syntax, plus genuinely neural typed-body search.
- Disjoint or constrained self/world heads and counterfactual interventions that break the current attribution gauge.
- Odrzywolek-style neural bootstrap tests that distinguish formula search, known-structure optimization, and sampling.
- External weakness studies whose group, task, and intervention define a real causal transport test rather than a correlational proxy.

17. Final synthesis

The complete system can be stated compactly.

1. Tasks induce equivalence relations on realizations.
2. Quotients retain task-relevant structure; fibers retain ignored variation.
3. Compilers and search processes determine which fiber members are accessible.

4. Concern reweights realization and action without automatically changing the quotient.
5. Viability and intervention close the loop from representation to control.
6. Representations form a task-relative Pareto antichain, not one hierarchy.
7. Obstruction pairs and cocycle discrepancies diagnose why a claim cannot descend or glue.
8. Repair restores a coordinate, changes a quotient, or adds transport, but no cheap universal repair selector survives.
9. Identification and admission should distinguish recovery, impossibility, budget limits, and next action.
10. Scientific discovery is credible only when residual content survives controls and changes an accepted type, operation, mechanism, or verifier.

The system's durable center is therefore not the slogan that geometry explains intelligence. It is a disciplined calculus of what a task treats as the same, what a process can realize, what an intervention can distinguish, what a system must preserve, and which claims survive their gates.

The grand synthesis remains a conjecture. The finite formal core is real. The empirical boundaries are unusually informative. The failures are not debris; they are the map of where the current regime stops.

Source map for the narrative

Primary synthesis sources include `README.md`, `TODO.md`, `docs/system_design.md`, `docs/discovery_regime_audit.md`, `docs/verification.md`, `docs/verification.json`, `docs/claim_registry.json`, `docs/program_evidence_registry.json`, `docs/experiment_contract_registry.json`, `docs/lea/theorem_backlog.md`, `docs/lea/VERIFY_RECEIPT_2026-08-17.md`, and `docs/lea/VERIFY_RECEIPT_2026-08-18.md`.

Core theory sources include `papers/structural_intelligence/paper.md`, `papers/structural_intelligence_foundations/paper.md`, `papers/theory_atlas/paper.md`, `papers/causal_semantics/paper.md`, `papers/representation_repair_calculus/paper.md`, `papers/delete_the_absolute/paper.md`, `papers/sic_dynamics/paper.md`, `papers/eml_universal_substrate/paper.md`, `papers/eml_access_geometry/paper.md`, `papers/concern_as_fiber_geometry/paper.md`, `papers/weakness_invariance_neurips/paper.md`, `papers/information_limited_discovery/paper.md`, `papers/relative_identifiability/paper.md`, and `papers/obstruction_aware_admission/paper.md`.

The generated appendices that follow in the PDF are authoritative indexes of the checked-in claim registry, evidence registry, experiment packages, preregistered hypotheses, result-status signals, and Lean declarations at the audited commit. They are generated from repository sources at build time so the atlas can be reproduced and drift can be detected.

Appendix A - Registered claims and evidence

These are the repository's centrally structured claims. A supported counterexample instrument can reject the broad claim it tests; local runner success and claim status are separate.

Claim ID	Tier / status	Statement	Evidence
MATH_RHO_KKT	theoretical / supported	For positive concern weights and an active capacity budget, the positive-price KKT solution scales as $\rho^*(x)$ proportional to $w(x)^{(d/(d+2))}$.	EVID-PRIMER-MATH-RHO
MATH_VOI_REDUCTION	descriptive / supported	Value of information is expected error reduction, MAE_{now} minus MAE_{after} , not the reverse difference.	EVID-PRIMER-MATH-VOI
COMMITMENT_GENERATOR_GENERALIZATION	internal / rejected	The frozen E5 design establishes generator-level generalization beyond labeled coverage.	EVID-COMMITMENT-E5-COVERAGE
COMMITMENT_E6_REPLICATION	internal / inconclusive	The E6 confirmatory replication is not adjudicated because its frozen smoke gate is blocked.	EVID-COMMITMENT-E6-SMOKE
WEAKNESS_EXTERNAL_PORTABILITY	external / rejected	Weakness-to-generalization causality transfers to external models under the current external-contact design.	EVID-EXTERNAL-WEAKNESS-P1
PASSIVE_ACTIVE_ATTRACTOR	causal / open	Passive-to-active results support a load-bearing, perturbation-resistant controller but do not establish an attractor basin or return dynamics.	EVID-PASSIVE-ACTIVE-ATTRACTOR
MATH_ASSUMPTION_DEPENDENCE	theoretical / supported	The seven registered theorem-bridge assumption families have executable satisfying examples and paired cases where the named assumption or bridge predicate fails; this audit does not establish theorem necessity, conclusion invalidation, or minimality.	EVID-MATH-ASSUMPTION-MATRIX
MATH_VOI_SURROGATE_GENERALITY	theoretical / rejected	Current error, posterior variance, error squared over K plus one, or mutual information universally rank probes by true expected regret reduction.	EVID-BAYESIAN-VOI-MISSPEC
SCIENCE_THREE_SEED_PROMOTION	internal / rejected	Three independent seeds provide promotion-ready uncertainty across the registered independent, hierarchical, null, and high-noise regimes.	EVID-SEED-BOOTSTRAP-S022
PHIL_GRADED_CAUSAL_USE	internal / supported	The registered mass-normalized dose-response score distinguishes a known causal feature from equally decodable-only and null features across synthetic commitment surfaces.	EVID-GRADED-CASUAL-USE
SYSTEMS_PASSIVE_ACTIVE_BIFURCATION	internal / rejected	Action coupling produces a reproducible discontinuous bifurcation across the registered order parameters, architectures, and seeds.	EVID-PASSIVE-ACTIVE-BIFURCATION
SYSTEMS_PASSIVE_ACTIVE_HYSTERESIS	internal / supported	Matched-budget forward and reverse continuation exhibit registered path dependence that survives washout and clears reinitialization controls in the synthetic passive-active model.	EVID-PASSIVE-ACTIVE-HYSTERESIS

Evidence records

Evidence ID	Status	Notes
EVID-PRIMER-MATH-RHO	pass	The corrected positive-price Lagrangian and $d=1$ allocation pass executable regression checks.
EVID-PRIMER-MATH-VOI	pass	Expected error reduction is now $MAE_{now} - MAE_{after}$; the heuristic is explicitly scoped.
EVID-COMMITMENT-E5-COVERAGE	fail	The frozen design fails generator, group-specificity, and transport gates; the apparent win is coverage-localized.
EVID-COMMITMENT-E6-SMOKE	inconclusive	Only 8 of 104 candidates meet the two-surface threshold versus 52 required; confirmatory execution is blocked.
EVID-EXTERNAL-WEAKNESS-P1	fail	The external-contact result does not support portable weakness-to-generalization causality.
EVID-PASSIVE-ACTIVE-ATTRACTOR	inconclusive	Load-bearing and perturbation-resistance proxies are present, but basin geometry and return dynamics were not measured.
EVID-MATH-ASSUMPTION-MATRIX	pass	All seven assumption families have deterministic satisfying examples and paired assumption/predicate failures. The audit does not test theorem necessity, downstream conclusion invalidation, or minimality.
EVID-BAYESIAN-VOI-MISSPEC	pass	Exact enumeration preserves the counterexample: the highest-information misspecified probe has zero oracle EVSI and negative true expected regret reduction, while a weaker robust signal is useful.
EVID-SEED-BOOTSTRAP-S022	pass	Three seeds were insufficient in every registered regime; promotion required 16 seeds in independent regimes, 64 in hierarchical regimes, and remained unavailable for the weak high-noise regime.
EVID-GRADED-CASUAL-USE	pass	A transparent synthetic SCM gives the policy-causal and decodable-only features identical perfect decoding accuracy while only the policy parent produces a positive mass-normalized dose response across two commitment surfaces; separate fixtures verify scale normalization.
EVID-PASSIVE-ACTIVE-BIFURCATION	fail	Only causal specificity favored a segmented fit; critical-point stability and multi-order-parameter co-location failed, so bifurcation is not supported. No separate positive smooth-crossover criterion was preregistered.
EVID-PASSIVE-ACTIVE-HYSTERESIS	pass	A seed-cluster bootstrap preserving both architectures yielded loop area 0.04471 across six contiguous significant continuation couplings and four under washout; the effect survived washout and cleared reinitialization controls.

Appendix B - Exhaustive experiment package ledger

The verification index contains 115 provenance-bearing packages. Entries report evidence plumbing, not automatic scientific endorsement. Legacy exceptions and partial histories are called out explicitly.

abstraction_frontier_pair

Package	abstraction_frontier_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/abstraction_frontier_pair/results/abstraction_frontier_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py abstraction_frontier_pair

activation_geometry

Package	activation_geometry
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/activation_geometry/results/alias_trained_behavior_direction_2026_06_08.md, experiments/activation_geometry/results/answer_surface_basin_diagnostic_2026_06_08.md, experiments/activation_geometry/results/attractor_pocket_diagnostic_2026_06_08.md, experiments/activation_geometry/results/behavior_alignment_breakthrough_2026_06_08.md, experiments/activation_geometry/results/behavior_level_gate_2026_06_08.md, experiments/activation_geometry/results/binary_layer3_replication_2026_06_10.md, experiments/activation_geometry/results/binary_layer3_scale_sweep_2026_06_10.md, experiments/activation_geometry/results/binary_pair_optimized_intervention_2026_06_10.md, experiments/activation_geometry/results/binary_pc_residualization_2026_06_10.md, experiments/activation_geometry/results/binary_pythia160_pocket_replication_2026_06_10.md, experiments/activation_geometry/results/binary_relation_behavior_gate_2026_06_09.md, experiments/activation_geometry/results/binary_yes_bias_controls_2026_06_09.md, experiments/activation_geometry/results/caa_cav_baseline_random_nulls_2026_06_09.md, experiments/activation_geometry/results/causal_patching_diagnostic_2026_06_08.md, experiments/activation_geometry/results/constrained_alias_direction_2026_06_09.md, experiments/activation_geometry/results/contrastive_binary_specificity_2026_06_09.md, experiments/activation_geometry/results/direction_subspace_diagnostic_2026_06_09.md, experiments/activation_geometry/results/feature_mask_binary_direction_2026_06_10.md, experiments/activation_geometry/results/final_token_steering_pilot_2026_06_08.md, experiments/activation_geometry/results/full_label_alias_gate_2026_06_08.md, experiments/activation_geometry/results/full_label_carrier_controls_2026_06_08.md, experiments/activation_geometry/results/generation_match_random_nulls_2026_06_09.md, experiments/activation_geometry/results/generation_readout_random_nulls_2026_06_09.md, experiments/activation_geometry/results/heldout_control_conditional_gate_2026_06_12.md, experiments/activation_geometry/results/heldout_readout_pilot_2026_06_08.md, experiments/activation_geometry/results/hook_output_dose_response_replication_2026_06_08.md, experiments/activation_geometry/results/intervention_layer_candidates_2026_06_08.md, experiments/activation_geometry/results/label_free_readout_basin_2026_06_08.md, experiments/activation_geometry/results/label_free_transfer_baseline_2026_06_08.md, experiments/activation_geometry/results/label_free_transfer_dose_response_2026_06_08.md, experiments/activation_geometry/results/matched_context_patching_2026_06_08.md, experiments/activation_geometry/results/modal_gpt2_layer_sweep_2026_06_08.md, experiments/activation_geometry/results/modal_pythia_70m_layer_last_2026_06_08.md, experiments/activation_geometry/results/modal_pythia_70m_layer_sweep_2026_06_08.md, experiments/activation_geometry/results/multialias_expanded_specificity_2026_06_09.md, experiments/activation_geometry/results/multiclass_state_gate_2026_06_10.md, experiments/activation_geometry/results/multiclass_state_gate_stress_2026_06_12.md, experiments/activation_geometry/results/multicontrol_alias_holdout_2026_06_09.md, experiments/activation_geometry/results/nightly_checkpoint_2026_06_10.md, experiments/activation_geometry/results/pair_control_diagnostic_2026_06_08.md, experiments/activation_geometry/results/pooling_ablation_pythia_gpt2_2026_06_08.md, experiments/activation_geometry/results/positive_family_binary_direction_2026_06_10.md, experiments/activation_geometry/results/positive_family_replication_2026_06_10.md, experiments/activation_geometry/results/pythia_matched_context_replication_2026_06_08.md, experiments/activation_geometry/results/random_relation_null_controls_2026_06_09.md, experiments/activation_geometry/results/readout_span_binary_direction_2026_06_10.md, experiments/activation_geometry/results/relation_control_state_gate_2026_06_10.md, experiments/activation_geometry/results/residualized_alias_direction_2026_06_08.md, experiments/activation_geometry/results/same_layer_hook_surface_diagnostic_2026_06_08.md, experiments/activation_geometry/results/short_answer_behavior_gate_2026_06_09.md, experiments/activation_geometry/results/state_gate_binary_direction_2026_06_10.md, experiments/activation_geometry/results/state_gate_robustness_2026_06_10.md, experiments/activation_geometry/results/steering_calibration_diagnostic_2026_06_08.md, experiments/activation_geometry/results/steering_gradient_alignment_2026_06_08.md, experiments/activation_geometry/results/stratified_binary_control_gate_2026_06_10.md, experiments/activation_geometry/results/target_disjoint_null_controls_2026_06_09.md, experiments/activation_geometry/results/target_family_pair_gate_2026_06_12.md, experiments/activation_geometry/results/trained_readout_gate_2026_06_08.md
Paper	No paper indexed
Reproduction	python scripts/regen.py activation_geometry

alignment_governance_pair

Package	alignment_governance_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/alignment_governance_pair/results/alignment_governance_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py alignment_governance_pair

allostatic_control

Package	allostatic_control
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/allostatic_control/paper.md
Reproduction	python scripts/regen.py allostatic_control

antecedent_taxonomy_pair

Package	antecedent_taxonomy_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/antecedent_taxonomy_pair/results/antecedent_taxonomy_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py antecedent_taxonomy_pair

autocatalytic_artwork_pair

Package	autocatalytic_artwork_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/autocatalytic_artwork_pair/results/autocatalytic_artwork_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py autocatalytic_artwork_pair

autopoietic_control

Package	autopoietic_control
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/autopoietic_control/paper.md
Reproduction	python scripts/regen.py autopoietic_control

bayesian_voi

Package	bayesian_voi
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=rejected
Result reports	experiments/bayesian_voi/results/bayesian_voi_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py bayesian_voi

causal_semantics_pair

Package	causal_semantics_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/causal_semantics_pair/results/causal_semantics_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py causal_semantics_pair

choice_dividend

Package	choice_dividend
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/choice_dividend/results/choice_dividend_summary.json
Paper	papers/choice_dividend/paper.md
Reproduction	python scripts/regen.py choice_dividend

commitment_surface

Package	commitment_surface
Evidence state	provenance=paper coverage=structured_manifest run=partial integrity=valid manifest=rejected adjudication=unadjudicated
Result reports	experiments/commitment_surface/results/m5_suite_c_reopen_reset_trigger_2026_07_14.json, experiments/commitment_surface/results/m5_suite_c_reopen_reset_trigger_2026_07_14.md
Paper	papers/commitment_surface/paper.md
Reproduction	python scripts/regen.py commitment_surface

compiler_tomography_pair

Package	compiler_tomography_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/compiler_tomography_pair/results/compiler_tomography_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py compiler_tomography_pair

concept_geometry

Package	concept_geometry
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/concept_geometry/results/openai_embedding_probe_2026_06_08.md, experiments/concept_geometry/results/paraphrase_stability_openai_2026_06_08.md
Paper	No paper indexed
Reproduction	python scripts/regen.py concept_geometry

concern_bootstrap

Package	concern_bootstrap
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/concern_bootstrap/paper.md
Reproduction	python scripts/regen.py concern_bootstrap

concern_fisher_pair

Package	concern_fisher_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/concern_fisher_pair/results/concern_fisher_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py concern_fisher_pair

concern_gated_retrieval

Package	concern_gated_retrieval
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/concern_gated_retrieval/results/summary.json, experiments/concern_gated_retrieval/results/pilot_2026_07_23.md
Paper	No paper indexed
Reproduction	python scripts/regen.py concern_gated_retrieval

concern_gated_retrieval_e2

Package	concern_gated_retrieval_e2
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/concern_gated_retrieval_e2/wave0/results/calibration_summary.json, experiments/concern_gated_retrieval_e2/wave0/PROVENANCE.md
Paper	No paper indexed
Reproduction	python scripts/regen.py concern_gated_retrieval_e2

concerned_syntax

Package	concerned_syntax
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/concerned_syntax/results/discovered_semantic_profiles_local_2026_06_22.md, experiments/concerned_syntax/results/discovered_semantic_profiles_modal_2026_06_22.md, experiments/concerned_syntax/results/intervention_invention_local_2026_06_16.md, experiments/concerned_syntax/results/intervention_invention_modal_2026_06_16.md, experiments/concerned_syntax/results/intervention_transfer_local_2026_06_16.md, experiments/concerned_syntax/results/intervention_transfer_modal_2026_06_16.md, experiments/concerned_syntax/results/intervention_transfer_repair_modal_2026_06_17.md, experiments/concerned_syntax/results/learned_agents_modal_2026_06_16.md, experiments/concerned_syntax/results/learned_object_slots_local_2026_06_22.md, experiments/concerned_syntax/results/learned_object_slots_modal_2026_06_22.md, experiments/concerned_syntax/results/learned_pixel_extractor_local_2026_06_17.md, experiments/concerned_syntax/results/learned_pixel_extractor_modal_2026_06_17.md, experiments/concerned_syntax/results/learned_slot_semantics_local_2026_06_18.md, experiments/concerned_syntax/results/learned_slot_semantics_modal_2026_06_18.md, experiments/concerned_syntax/results/mechanism_trace_modal_2026_06_17.md, experiments/concerned_syntax/results/modal_sweep_2026_06_16.md, experiments/concerned_syntax/results/pilot_2026_06_16.md, experiments/concerned_syntax/results/pixel_shapes_local_2026_06_16.md, experiments/concerned_syntax/results/pixel_shapes_modal_2026_06_16.md, experiments/concerned_syntax/results/rich_program_language_local_2026_06_17.md, experiments/concerned_syntax/results/rich_program_language_modal_2026_06_17.md, experiments/concerned_syntax/results/rich_program_transfer_repair_local_2026_06_18.md, experiments/concerned_syntax/results/rich_program_transfer_repair_modal_2026_06_18.md, experiments/concerned_syntax/results/searched_program_policy_modal_2026_06_17.md, experiments/concerned_syntax/results/searched_rich_program_policy_local_2026_06_18.md, experiments/concerned_syntax/results/searched_rich_program_policy_modal_2026_06_18.md, experiments/concerned_syntax/results/unsupervised_slot_semantics_local_2026_06_18.md, experiments/concerned_syntax/results/unsupervised_slot_semantics_modal_2026_06_18.md, experiments/concerned_syntax/results/vector_shapes_local_2026_06_16.md, experiments/concerned_syntax/results/vector_shapes_modal_2026_06_16.md
Paper	papers/concerned_syntax/paper.md
Reproduction	python scripts/regen.py concerned_syntax

constraint_swap_causal_geometry

Package	constraint_swap_causal_geometry
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=rejected adjudication=unadjudicated
Result reports	experiments/constraint_swap_causal_geometry/results/summary.json, experiments/constraint_swap_causal_geometry/results/summary.md, papers/constraint_swap_causal_geometry/paper.pdf
Paper	papers/constraint_swap_causal_geometry/paper.md
Reproduction	python scripts/regen.py constraint_swap_causal_geometry

copernicus_geocentrism_corpus

Package	copernicus_geocentrism_corpus
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/copernicus_geocentrism_corpus/results/fetch_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py copernicus_geocentrism_corpus

costly_null_probes

Package	costly_null_probes
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/costly_null_probes/paper.md
Reproduction	python scripts/regen.py costly_null_probes

cross_task_learnability

Package	cross_task_learnability
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/cross_task_learnability/results/cross_task_learnability_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py cross_task_learnability

cross_task_learnability_continuous

Package	cross_task_learnability_continuous
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/cross_task_learnability_continuous/results/cross_task_learnability_continuous_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py cross_task_learnability_continuous

cross_task_sufficiency

Package	cross_task_sufficiency
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/cross_task_sufficiency/results/cross_task_sufficiency_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py cross_task_sufficiency

current_error_calibration

Package	current_error_calibration
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/current_error_calibration/paper.md
Reproduction	python scripts/regen.py current_error_calibration

darwin_species_fixity_corpus

Package	darwin_species_fixity_corpus
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/darwin_species_fixity_corpus/results/fetch_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py darwin_species_fixity_corpus

date_cut_retrodition

Package	date_cut_retrodition
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/date_cut_retrodition/results/dcr1_verdict.json
Paper	No paper indexed
Reproduction	python scripts/regen.py date_cut_retrodition

delete_repair_concern

Package	delete_repair_concern
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_concern/results/delete_repair_concern_summary.json
Paper	papers/delete_repair_concern/paper.md
Reproduction	python scripts/regen.py delete_repair_concern

delete_repair_concern_estimation

Package	delete_repair_concern_estimation
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_concern_estimation/results/delete_repair_concern_estimation_summary.json
Paper	papers/delete_repair_concern_estimation/paper.md
Reproduction	python scripts/regen.py delete_repair_concern_estimation

delete_repair_concern_transport

Package	delete_repair_concern_transport
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_concern_transport/results/delete_repair_concern_transport_summary.json
Paper	papers/delete_repair_concern_transport/paper.md
Reproduction	python scripts/regen.py delete_repair_concern_transport

delete_repair_connection

Package	delete_repair_connection
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_connection/results/delete_repair_connection_summary.json
Paper	papers/delete_repair_connection/paper.md
Reproduction	python scripts/regen.py delete_repair_connection

delete_repair_disanalogy

Package	delete_repair_disanalogy
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_disanalogy/results/delete_repair_disanalogy_summary.json
Paper	papers/delete_repair_disanalogy/paper.md
Reproduction	python scripts/regen.py delete_repair_disanalogy

delete_repair_generators

Package	delete_repair_generators
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_generators/results/delete_repair_generators_summary.json
Paper	papers/delete_repair_generators/paper.md
Reproduction	python scripts/regen.py delete_repair_generators

delete_repair_kappa

Package	delete_repair_kappa
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_kappa/results/delete_repair_kappa_summary.json
Paper	papers/delete_repair_kappa/paper.md
Reproduction	python scripts/regen.py delete_repair_kappa

delete_repair_menu_blind

Package	delete_repair_menu_blind
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_menu_blind/results/delete_repair_menu_blind_summary.json
Paper	papers/delete_repair_menu_blind/paper.md
Reproduction	python scripts/regen.py delete_repair_menu_blind

delete_repair_reduction

Package	delete_repair_reduction
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_reduction/results/delete_repair_reduction_summary.json
Paper	papers/delete_repair_reduction/paper.md
Reproduction	python scripts/regen.py delete_repair_reduction

delete_repair_surgery

Package	delete_repair_surgery
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_surgery/results/delete_repair_surgery_summary.json
Paper	papers/delete_repair_surgery/paper.md
Reproduction	python scripts/regen.py delete_repair_surgery

delete_repair_swap

Package	delete_repair_swap
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_repair_swap/results/delete_repair_swap_summary.json
Paper	papers/delete_repair_swap/paper.md
Reproduction	python scripts/regen.py delete_repair_swap

delete_the_absolute

Package	delete_the_absolute
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/delete_the_absolute/results/delete_the_absolute_summary.json
Paper	papers/delete_the_absolute/paper.md
Reproduction	python scripts/regen.py delete_the_absolute

deletion_repair

Package	deletion_repair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/deletion_repair/results/dr1_verdict.json
Paper	No paper indexed
Reproduction	python scripts/regen.py deletion_repair

dial_nestedness

Package	dial_nestedness
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/dial_nestedness/results/dial_nestedness_summary.json
Paper	papers/dial_nestedness/paper.md
Reproduction	python scripts/regen.py dial_nestedness

dr6_code_correctness_corollary

Package	dr6_code_correctness_corollary
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/dr6_code_correctness_corollary/results/dr6_verdict.json
Paper	No paper indexed
Reproduction	python scripts/regen.py dr6_code_correctness_corollary

eml_fiber_spectrum

Package	eml_fiber_spectrum
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/eml_fiber_spectrum/results/eml_fiber_spectrum_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py eml_fiber_spectrum

eml_us4_discrete

Package	eml_us4_discrete
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/eml_us4_discrete/results/eml_us4_discrete_summary.json
Paper	papers/eml_us4_discrete/paper.md
Reproduction	python scripts/regen.py eml_us4_discrete

eml_us4_gradient

Package	eml_us4_gradient
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/eml_us4_gradient/results/eml_us4_gradient_summary.json
Paper	papers/eml_us4_gradient/paper.md
Reproduction	python scripts/regen.py eml_us4_gradient

eml_us4_prime

Package	eml_us4_prime
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/eml_us4_prime/results/eml_us4_prime_summary.json
Paper	papers/eml_us4_prime/paper.md
Reproduction	python scripts/regen.py eml_us4_prime

eml_us4_search

Package	eml_us4_search
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/eml_us4_search/results/eml_us4_search_summary.json
Paper	papers/eml_us4_search/paper.md
Reproduction	python scripts/regen.py eml_us4_search

eml_variable_spectrum

Package	eml_variable_spectrum
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/eml_variable_spectrum/results/eml_variable_spectrum_summary.json
Paper	papers/eml_variable_spectrum/paper.md
Reproduction	python scripts/regen.py eml_variable_spectrum

ensemble_uncertainty

Package	ensemble_uncertainty
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/ensemble_uncertainty/paper.md
Reproduction	python scripts/regen.py ensemble_uncertainty

epistemic_exploration

Package	epistemic_exploration
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/epistemic_exploration/paper.md
Reproduction	python scripts/regen.py epistemic_exploration

exploration_diagnostics

Package	exploration_diagnostics
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/exploration_diagnostics/paper.md
Reproduction	python scripts/regen.py exploration_diagnostics

external_contact

Package	external_contact
Evidence state	provenance=results coverage=structured_manifest run=partial integrity=valid manifest=rejected adjudication=rejected
Result reports	experiments/external_contact/results/p1_pythia_lora_2026_06_22.md
Paper	No paper indexed
Reproduction	python scripts/regen.py external_contact

first_order_self

Package	first_order_self
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/first_order_self/paper.md
Reproduction	python scripts/regen.py first_order_self

future_commitment_quotient

Package	future_commitment_quotient
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/future_commitment_quotient/claim_calibration.json, experiments/future_commitment_quotient/results/registered_rows.jsonl, experiments/future_commitment_quotient/results/summary.json, experiments/future_commitment_quotient/results/summary.md, papers/future_commitment_quotient/paper.pdf
Paper	papers/future_commitment_quotient/paper.md
Reproduction	python scripts/regen.py future_commitment_quotient

gauge_fixed_concern_transport

Package	gauge_fixed_concern_transport
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/gauge_fixed_concern_transport/results/gfc_l4_suite_2026_07_07.json, experiments/gauge_fixed_concern_transport/results/gfc_l4_suite_2026_07_07.md, experiments/gauge_fixed_concern_transport/results/gfc_local_full_rerun_2026_07_08.json, experiments/gauge_fixed_concern_transport/results/gfc_local_full_rerun_2026_07_08.md, experiments/gauge_fixed_concern_transport/results/gfc_smoke_suite_2026_07_07.json, experiments/gauge_fixed_concern_transport/results/gfc_smoke_suite_2026_07_07.md
Paper	papers/gauge_fixed_concern_transport/paper.md
Reproduction	python scripts/regen.py gauge_fixed_concern_transport

grid_cell_weakness

Package	grid_cell_weakness
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/grid_cell_weakness/results/capacity_bottleneck_2026_07_01.md, experiments/grid_cell_weakness/results/grid_cell_weakness_conference_evidence_2026_07_02.md, experiments/grid_cell_weakness/results/local_cpu_sweep_2026_06_29.md, experiments/grid_cell_weakness/results/modal_grid_cell_weakness_sweep_2026_07_02.md, experiments/grid_cell_weakness/results/pilot_2026_06_28.md, experiments/grid_cell_weakness/results/ratedistortion_2026_07_01.md, experiments/grid_cell_weakness/results/reward_deformation_2026_06_29.md, experiments/grid_cell_weakness/results/reward_deformation_sweep_2026_07_02.md, experiments/grid_cell_weakness/results/reward_location_sweep_2026_07_02.md
Paper	No paper indexed
Reproduction	python scripts/regen.py grid_cell_weakness

grounded_statecharts

Package	grounded_statecharts
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/grounded_statecharts/results/harness_unlearning/summary.json, experiments/grounded_statecharts/results/harness_unlearning/replay.html
Paper	No paper indexed
Reproduction	python scripts/regen.py grounded_statecharts

habituated_reengagement

Package	habituated_reengagement
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/habituated_reengagement/paper.md
Reproduction	python scripts/regen.py habituated_reengagement

homeostatic_objects

Package	homeostatic_objects
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/homeostatic_objects/paper.md
Reproduction	python scripts/regen.py homeostatic_objects

ident

Package	ident
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/ident/results/baseline_summary.json, experiments/ident/results/baseline_summary.md, experiments/ident/results/model_summary.json, experiments/ident/results/model_summary.md
Paper	papers/ident/paper.md
Reproduction	python scripts/regen.py ident

information_limited_discovery

Package	information_limited_discovery
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/information_limited_discovery/results/summary.json, experiments/information_limited_discovery/results/summary.md, papers/information_limited_discovery/paper.md, papers/information_limited_discovery/paper.pdf
Paper	papers/information_limited_discovery/paper.md
Reproduction	python scripts/regen.py information_limited_discovery

interventional_contrast

Package	interventional_contrast
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/interventional_contrast/paper.md
Reproduction	python scripts/regen.py interventional_contrast

interventional_crl_learnability

Package	interventional_crl_learnability
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/interventional_crl_learnability/results/interventional_crl_learnability_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py interventional_crl_learnability

ivae_learnability

Package	ivae_learnability
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/ivae_learnability/results/ivae_learnability_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py ivae_learnability

lavoisier_phlogiston_corpus

Package	lavoisier_phlogiston_corpus
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/lavoisier_phlogiston_corpus/results/fetch_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py lavoisier_phlogiston_corpus

learned_symmetry

Package	learned_symmetry
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/learned_symmetry/results/causal_v1_2026_06_09.md, experiments/learned_symmetry/results/modal_sweep_v1_2026_06_09.md
Paper	No paper indexed
Reproduction	python scripts/regen.py learned_symmetry

linear_ica_learnability

Package	linear_ica_learnability
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/linear_ica_learnability/results/linear_ica_learnability_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py linear_ica_learnability

load_bearing_prose_test

Package	load_bearing_prose_test
Evidence state	provenance=results coverage=structured_manifest run=partial integrity=valid manifest=planned adjudication=unadjudicated
Result reports	experiments/load_bearing_prose_test/results/summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py load_bearing_prose_test

long_horizon_bottleneck

Package	long_horizon_bottleneck
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/long_horizon_bottleneck/results/modal_transformer_l4_8seed_2026_07_02.md, experiments/long_horizon_bottleneck/results/modal_transformer_l4_horizon_4seed_2026_07_02.md, experiments/long_horizon_bottleneck/results/tool_commitment_l4_4seed_2026_07_02.md, experiments/long_horizon_bottleneck/results/z_closed_loop_tool_commitment_l4_4seed_2026_07_02.md, experiments/long_horizon_bottleneck/results/zz_structured_tool_call_local_smoke_2026_07_03.md, experiments/long_horizon_bottleneck/results/zz_tool_recovery_l4_4seed_2026_07_02.md, experiments/long_horizon_bottleneck/results/zzz_structured_tool_call_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzz_multifield_tool_schema_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzz_stochastic_tool_failure_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzz_stochastic_tool_failure_8slot_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzz_alias_argument_surface_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzz_text_argument_surface_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzzz_generated_json_surface_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzz_autoregressive_json_surface_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzz_prompt_json_transfer_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzzz_prompt_json_hidden_localization_smoke_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzzzz_prompt_json_hidden_localization_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzzzzz_prompt_json_fixed_action_localization_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzzzzzz_prompt_json_causal_patch_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzzzzzzz_prompt_json_prompt_family_causal_patch_l4_4seed_2026_07_03.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzzzzzzzzz_api_blackbox_gemini_flash_lite_2026_07_06.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzzzzzzzzzz_api_blackbox_multi_provider_2026_07_06.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzzzzzzzzzzz_api_dispatch_characterization_openai_gpt41_nano_2026_07_06.md, experiments/long_horizon_bottleneck/results/zzzzzzzzzzzzzzzzzzzzz_api_dispatch_robustness_openai_gpt41_nano_2026_07_06.md
Paper	papers/long_horizon_bottleneck/paper.md
Reproduction	python scripts/regen.py long_horizon_bottleneck

mathematical_claims

Package	mathematical_claims
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=supported
Result reports	experiments/mathematical_claims/results/mathematical_claims_summary.json, experiments/mathematical_claims/theorem_assumption_matrix.json
Paper	No paper indexed
Reproduction	python scripts/regen.py mathematical_claims

neural_group_generator

Package	neural_group_generator
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/neural_group_generator/results/cluttered_mnist_2026_06_09.md, experiments/neural_group_generator/results/negative_results_2026_06_09.md, experiments/neural_group_generator/results/rotated_mnist_2026_06_09.md, experiments/neural_group_generator/results/topk_ablation_2026_06_10.md
Paper	papers/neural_group_generator/paper.md
Reproduction	python scripts/regen.py neural_group_generator

null_intervention

Package	null_intervention
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/null_intervention/paper.md
Reproduction	python scripts/regen.py null_intervention

obstruction_aware_admission

Package	obstruction_aware_admission
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/obstruction_aware_admission/results/summary.json, experiments/obstruction_aware_admission/results/summary.md, papers/obstruction_aware_admission/paper.md, papers/obstruction_aware_admission/paper.pdf
Paper	papers/obstruction_aware_admission/paper.md
Reproduction	python scripts/regen.py obstruction_aware_admission

off_policy_state_coverage

Package	off_policy_state_coverage
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/off_policy_state_coverage/paper.md
Reproduction	python scripts/regen.py off_policy_state_coverage

online_identifying_interventions

Package	online_identifying_interventions
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/online_identifying_interventions/paper.md
Reproduction	python scripts/regen.py online_identifying_interventions

pac_bayes_weakness_enum

Package	pac_bayes_weakness_enum
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/pac_bayes_weakness_enum/results/pac_bayes_weakness_enum_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py pac_bayes_weakness_enum

paraphrase_weakness

Package	paraphrase_weakness
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/paraphrase_weakness/results/learned_substitution_2026_06_09.md, experiments/paraphrase_weakness/results/pythia_70m_2026_06_09.md
Paper	No paper indexed
Reproduction	python scripts/regen.py paraphrase_weakness

passive_active_phase_map

Package	passive_active_phase_map
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=rejected adjudication=rejected, supported
Result reports	experiments/passive_active_phase_map/results/registered_summary.json, experiments/passive_active_phase_map/results/registered_summary.md
Paper	No paper indexed
Reproduction	python scripts/regen.py passive_active_phase_map

passive_to_active

Package	passive_to_active
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/passive_to_active/results/pythia_70m_v2_2026_06_10.md
Paper	No paper indexed
Reproduction	python scripts/regen.py passive_to_active

phase4_metaphysics

Package	phase4_metaphysics
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/phase4_metaphysics/results/phase4_l4_suite_2026_07_06.md
Paper	papers/phase4_metaphysics/paper.md
Reproduction	python scripts/regen.py phase4_metaphysics

phase5_external_validity

Package	phase5_external_validity
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/phase5_external_validity/results/phase5_l4_suite_2026_07_06.md
Paper	papers/phase5_external_validity/paper.md
Reproduction	python scripts/regen.py phase5_external_validity

phase6_real_model_validation

Package	phase6_real_model_validation
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/phase6_real_model_validation/results/l4_full_suite_2026_07_06.json, experiments/phase6_real_model_validation/results/phase6_real_model_suite_2026_07_06.md
Paper	papers/phase6_real_model_validation/paper.md
Reproduction	python scripts/regen.py phase6_real_model_validation

planning_from_concern

Package	planning_from_concern
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/planning_from_concern/paper.md
Reproduction	python scripts/regen.py planning_from_concern

planning_hardening

Package	planning_hardening
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/planning_hardening/paper.md
Reproduction	python scripts/regen.py planning_hardening

probe_value_reengagement

Package	probe_value_reengagement
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/probe_value_reengagement/paper.md
Reproduction	python scripts/regen.py probe_value_reengagement

rate_distortion_pair

Package	rate_distortion_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/rate_distortion_pair/results/rate_distortion_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py rate_distortion_pair

regime_sensitive_de

Package	regime_sensitive_de
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/regime_sensitive_de/paper.md
Reproduction	python scripts/regen.py regime_sensitive_de

relative_identifiability

Package	relative_identifiability
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/relative_identifiability/results/summary.json, experiments/relative_identifiability/results/summary.md, papers/relative_identifiability/paper.md
Paper	papers/relative_identifiability/paper.md
Reproduction	python scripts/regen.py relative_identifiability

representation_repair_pair

Package	representation_repair_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/representation_repair_pair/results/representation_repair_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py representation_repair_pair

representation_search

Package	representation_search
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/representation_search/results/representation_search_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py representation_search

role_specific_identifiability

Package	role_specific_identifiability
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/role_specific_identifiability/paper.md
Reproduction	python scripts/regen.py role_specific_identifiability

rotation_weakness

Package	rotation_weakness
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/rotation_weakness/results/sweep_v1_2026_06_09.md
Paper	No paper indexed
Reproduction	python scripts/regen.py rotation_weakness

scale_normalized_vprobe

Package	scale_normalized_vprobe
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/scale_normalized_vprobe/paper.md
Reproduction	python scripts/regen.py scale_normalized_vprobe

seed_bootstrap_calibration

Package	seed_bootstrap_calibration
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=rejected
Result reports	experiments/seed_bootstrap_calibration/results/summary.json, experiments/seed_bootstrap_calibration/results/summary.md
Paper	No paper indexed
Reproduction	python scripts/regen.py seed_bootstrap_calibration

semantic_concern_geometry

Package	semantic_concern_geometry
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/semantic_concern_geometry/results/semantic_concern_sweep_2026_07_02.md
Paper	No paper indexed
Reproduction	python scripts/regen.py semantic_concern_geometry

sica_finite_derivation_pair

Package	sica_finite_derivation_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/sica_finite_derivation_pair/results/sica_finite_derivation_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py sica_finite_derivation_pair

sicc_covering_meta_pair

Package	sicc_covering_meta_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/sicc_covering_meta_pair/results/sicc_covering_meta_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py sicc_covering_meta_pair

silent_substitution

Package	silent_substitution
Evidence state	provenance=paper coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/silent_substitution/results/silent_substitution_summary.json
Paper	papers/silent_substitution/paper.md
Reproduction	python scripts/regen.py silent_substitution

sparse_ica_learnability

Package	sparse_ica_learnability
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/sparse_ica_learnability/results/sparse_ica_learnability_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py sparse_ica_learnability

squaring_separation

Package	squaring_separation
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/squaring_separation/results/squaring_separation_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py squaring_separation

state_dependent_concern

Package	state_dependent_concern
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/state_dependent_concern/paper.md
Reproduction	python scripts/regen.py state_dependent_concern

structure_compatible_generalization

Package	structure_compatible_generalization
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/structure_compatible_generalization/results/language_template_substitution_2026_07_06.md, experiments/structure_compatible_generalization/results/phase2_transformations_2026_07_06.md, experiments/structure_compatible_generalization/results/phase3_learned_generators_2026_07_06.md, experiments/structure_compatible_generalization/results/phase_row_ledgers_2026_07_06.md, experiments/structure_compatible_generalization/results/phase_row_ledgers_manifest_2026_07_06.json, experiments/structure_compatible_generalization/results/semantic_retrieval_transfer_2026_07_06.md, experiments/structure_compatible_generalization/results/semantic_selection_bootstrap_2026_07_06.json, experiments/structure_compatible_generalization/results/semantic_selection_bootstrap_2026_07_06.md, experiments/structure_compatible_generalization/results/semantic_selection_control_2026_07_06.md, experiments/structure_compatible_generalization/results/semantic_selection_row_release_2026_07_06.md, experiments/structure_compatible_generalization/results/semantic_selection_row_release_manifest_2026_07_06.json, experiments/structure_compatible_generalization/results/semantic_selection_tiebreak_stress_2026_07_06.json, experiments/structure_compatible_generalization/results/semantic_selection_tiebreak_stress_2026_07_06.md, experiments/structure_compatible_generalization/results/structure_compatible_l4_2026_07_06.md
Paper	No paper indexed
Reproduction	python scripts/regen.py structure_compatible_generalization

structure_compiler

Package	structure_compiler
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/structure_compiler/results/structure_compiler_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py structure_compiler

symbolic_causation

Package	symbolic_causation
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/symbolic_causation/results/symbolic_causation_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py symbolic_causation

symbolic_weakness

Package	symbolic_weakness
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/symbolic_weakness/results/modal_neural_sweep_2026_07_02.md, experiments/symbolic_weakness/results/modal_neural_sweep_v1_2026_06_09.md, experiments/symbolic_weakness/results/multi_family_500_2026_06_09.md, experiments/symbolic_weakness/results/neural_correlation_checks_2026_07_02.md, experiments/symbolic_weakness/results/neural_sweep_v3_2026_06_09.md, experiments/symbolic_weakness/results/prefix_shift_pilot_2026_06_09.md
Paper	No paper indexed
Reproduction	python scripts/regen.py symbolic_weakness

theory_atlas_pair

Package	theory_atlas_pair
Evidence state	provenance=results coverage=structured_manifest run=complete integrity=valid manifest=accepted adjudication=unadjudicated
Result reports	experiments/theory_atlas_pair/results/theory_atlas_pair_summary.json
Paper	No paper indexed
Reproduction	python scripts/regen.py theory_atlas_pair

two_bottlenecks

Package	two_bottlenecks
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/two_bottlenecks/paper.md
Reproduction	python scripts/regen.py two_bottlenecks

valence_object_formation

Package	valence_object_formation
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/valence_object_formation/paper.md
Reproduction	python scripts/regen.py valence_object_formation

valence_tapestry

Package	valence_tapestry
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/valence_tapestry/paper.md
Reproduction	python scripts/regen.py valence_tapestry

vector_first_order_self

Package	vector_first_order_self
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	papers/vector_first_order_self/paper.md
Reproduction	python scripts/regen.py vector_first_order_self

viable_computational_bodies

Package	viable_computational_bodies
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/viable_computational_bodies/results/executable_bodies_modal_2026_06_16.md, experiments/viable_computational_bodies/results/learned_executable_modules_local_2026_06_18.md, experiments/viable_computational_bodies/results/learned_executable_modules_modal_2026_06_18.md, experiments/viable_computational_bodies/results/modal_sweep_2026_06_16.md, experiments/viable_computational_bodies/results/object_slot_executable_modules_local_2026_06_22.md, experiments/viable_computational_bodies/results/object_slot_executable_modules_modal_2026_06_22.md, experiments/viable_computational_bodies/results/pilot_2026_06_16.md, experiments/viable_computational_bodies/results/program_body_search_haskell_local_2026_06_16.md, experiments/viable_computational_bodies/results/program_body_search_modal_2026_06_16.md, experiments/viable_computational_bodies/results/rich_program_body_search_modal_2026_06_18.md, experiments/viable_computational_bodies/results/searched_executable_modules_local_2026_06_22.md, experiments/viable_computational_bodies/results/searched_executable_modules_modal_2026_06_22.md, experiments/viable_computational_bodies/results/vector_module_bodies_local_2026_06_16.md, experiments/viable_computational_bodies/results/vector_module_bodies_modal_2026_06_16.md
Paper	papers/viable_computational_bodies/paper.md
Reproduction	python scripts/regen.py viable_computational_bodies

virtual_governor_stress_signal

Package	virtual_governor_stress_signal
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/virtual_governor_stress_signal/results/virtual_governor_stress_signal_l4_2026_07_06.md
Paper	papers/virtual_governor_stress_signal/paper.md
Reproduction	python scripts/regen.py virtual_governor_stress_signal

weakness_temporal

Package	weakness_temporal
Evidence state	provenance=scaffold coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	No committed result report indexed
Paper	No paper indexed
Reproduction	python scripts/regen.py weakness_temporal

weakness_vs_simplicity

Package	weakness_vs_simplicity
Evidence state	provenance=results coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/weakness_vs_simplicity/results/broad_negative_excluder_stress_2026_06_08.md, experiments/weakness_vs_simplicity/results/pilot_2026_06_08.md, experiments/weakness_vs_simplicity/results/validated_weakness_stress_2026_06_08.md
Paper	No paper indexed
Reproduction	python scripts/regen.py weakness_vs_simplicity

world_responds

Package	world_responds
Evidence state	provenance=paper coverage=legacy_exception run=None integrity=None manifest=None adjudication=unadjudicated
Result reports	experiments/world_responds/results/reengagement_23a_2026_07_06.md, experiments/world_responds/results/suite_c_neural_transfer_2026_07_06.json, experiments/world_responds/results/suite_c_neural_transfer_2026_07_06.md, experiments/world_responds/results/suite_c_reengagement_2026_07_06.json, experiments/world_responds/results/suite_c_reengagement_2026_07_06.md, experiments/world_responds/results/suite_c_source_ablation_transfer_2026_07_06.json, experiments/world_responds/results/suite_c_source_ablation_transfer_2026_07_06.md, experiments/world_responds/results/suite_c_teacher_free_inquiry_2026_07_06.json, experiments/world_responds/results/suite_c_teacher_free_inquiry_2026_07_06.md, experiments/world_responds/results/suite_c_teacher_free_wide_stats_2026_07_06.json, experiments/world_responds/results/suite_c_teacher_free_wide_stats_2026_07_06.md
Paper	papers/world_responds/paper.md
Reproduction	python scripts/regen.py world_responds

Appendix C - Preregistered hypotheses and recorded outcomes

This appendix includes all 46 top-level JSON preregistrations. Paper-only and nested wave preregistrations remain visible in Appendix B through their packages and reports.

abstraction_frontier_pair

Hypothesis: Theorems AF-1 and AF-2 (paper papers/abstraction_frontier/paper.md) predict, on the 4-bit Boolean world with the shared-through-Z task family (parity{0,1}, parity{2,3}, parity{0,1,2,3}), that the Pareto frontier over the four axes (task-sufficiency = $\max_{\alpha} H(Y_{\alpha} | q(X))$, coding cost = $\log_2(\text{image}(q))$, dynamical closure = 0, control regret = 0) is a two-element antichain: the constant map (min-coding-cost, max-task-sufficiency endpoint) and the true $Z = \text{joint}(\text{parity}\{0,1\}, \text{parity}\{2,3\})$ (unique zero-task-sufficiency frontier member). The identity is dominated by Z (same sufficiency 0, strictly higher coding cost $4 > 2$) -- the static-case collapse of AF-2's 'segment from q^* to identity' statement.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: abstraction_frontier_pair_summary.json: gates 6/6 true; abstraction_frontier_pair_summary.json:status=pass

Source: experiments/abstraction_frontier_pair/preregistration.json

alignment_governance_pair

Hypothesis: Theorem AG-1 (paper papers/alignment_as_ensemble_governance/paper.md) predicts that on a 4-state Markov world with viable region $V=\{0,1,2\}$, unviable state $\{3\}$, uniform random policy over actions {stay, move}, and per-step leakage rate $\beta=0.05$ by construction, the survival probability $\Pr[q(X_t) \text{ in } V \text{ for all } t \leq T]$ is at least $(1-\beta)^T$ at every T in $\{1, 3, 5, 10, 20\}$, computed exactly by matrix powers. On this world the bound is tight (equality) at every T because every row of the V -sub-kernel sums to exactly $1-\beta$. Theorem AG-2 predicts extending V to $V'=Z$ gives survival 1.0 at every T (trivially, since $\beta_Z=0$).

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: alignment_governance_pair_summary.json: gates 4/4 true; alignment_governance_pair_summary.json:status=pass

Source: experiments/alignment_governance_pair/preregistration.json

antecedent_taxonomy_pair

Hypothesis: Theorem SA-1 (paper papers/sufficient_antecedents/paper.md) predicts that four canonical inductive-bias regimes (linear ICA, sparse-linear ICA, auxiliary-variable iVAE, interventional CRL) each satisfy Condition (I) local separation and Condition (II) cross-u coherence on the 4-bit Boolean world; the finest common refinement of each antecedent's local screens equals the latent Z -partition exactly (up to trivial refinement for the interventional case whose observational screen already gives Z).

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: antecedent_taxonomy_pair_summary.json: gates 4/4 true; antecedent_taxonomy_pair_summary.json:status=pass

Source: experiments/antecedent_taxonomy_pair/preregistration.json

autocatalytic_artwork_pair

Hypothesis: For the 6-step, 3-compiler autocatalytic-artwork world of the companion paper *Autocatalytic Artwork* (papers/autocatalytic_artwork/paper.md), with specification alphabet $|S|=4$, experience alphabet $|E|=4$, candidate compiler family $\{K_a \text{ (diagonal } 0.85/0.05), K_b \text{ (shifted } 0.85/0.05), K_c \text{ (uniform } 0.25)\}$, true compiler K_a , uniform prior $\mu_0=(1/3,1/3,1/3)$, and 200 seeded Mulberry32 runs, Theorem AA-1 predicts (a) the audience's mean analytical expected one-step predictive log-likelihood $LL_t = E_{\{s, e \sim P_S \times K_a\}}[\log K_t(e|s)]$ is non-decreasing in t from $t=0$ through $t=6$, and (b) the mean posterior mass on the true compiler K_a at $t=6$ is at least 0.9. Theorem AA-2 predicts (c) the Bayesian update $\mu_{\{t+1\}}(\theta)$ proportional to $\mu_t(\theta) K_{\theta}(e_{t+1}|s_t)$ is exactly the Boltzmann compiler-ecology update of Theorem CT-2 with reward $r_t(\theta)=\log K_{\theta}(e_{t+1}|s_t)$ and $\beta=1$, so a parallel Boltzmann trajectory agrees with the Bayesian trajectory to $1e-12$ on every (t, θ, seed) . Companion gate (d): at $t=6$ the Bayesian-audience expected LL exceeds the frozen uniform-baseline expected LL by at least 0.1 nats.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: autocatalytic_artwork_pair_summary.json: gates 4/4 true; autocatalytic_artwork_pair_summary.json:status=pass

Source: experiments/autocatalytic_artwork_pair/preregistration.json

bayesian_voi

Hypothesis: Exact expected value of sample information ranks informative probes better than current-error-only heuristics, while irreducible noise and misspecification expose their failure regimes.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: bayesian_voi_summary.json: gates 5/5 true; bayesian_voi_summary.json:status=pass

Source: experiments/bayesian_voi/preregistration.json

causal_semantics_pair

Hypothesis: Theorems CS-1 and CS-2 (paper papers/causal_semantics/paper.md) predict that for a message space $M=\{m_0,\dots,m_5\}$, a context space $C=\{c_0,\dots,c_3\}$, and the hand-built update operator $\Psi: M \times C \rightarrow P(X)$ with $X=\{x_0,\dots,x_3\}$ of paper section 4, (a) the binary relation $\sim_\Psi$ on M defined by ' $\Psi(m_1, c) == \Psi(m_2, c)$ for every c in C ' is a proper equivalence relation (reflexive, symmetric, transitive) on M ; (b) its quotient $M/\sim_\Psi$ is exactly the four-class partition $\{\{m_0,m_1\},\{m_2,m_3\},\{m_4\},\{m_5\}\}$; (c) $M/\sim_\Psi$ is a common sufficient statistic (in the Theorem-4 sense) for the family $\{\Psi(\cdot, c) : c \text{ in } C\}$ on M , meaning $\Psi(\cdot, c)$ is constant within each Ψ -class for every c ; and (d) the co-occurrence quotient $M/\sim_\kappa$ induced by the distractor signature κ (even-indexed messages $\{m_0,m_2,m_4\}$ share signature $(4,4,1,1)$; odd-indexed messages $\{m_1,m_3,m_5\}$ share signature $(1,1,4,4)$) is distinct from $M/\sim_\Psi$ -- neither partition refines the other and no cell coincides.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: causal_semantics_pair_summary.json: gates 4/4 true; causal_semantics_pair_summary.json:status=pass

Source: experiments/causal_semantics_pair/preregistration.json

choice_dividend

Hypothesis: Gate 1 of the essay 'Intention Is All You Need' (the dividend gate) holds at the kernel: on the sixteen worlds $\{0,1\}^4$ the choice dividend of a task (D12: best compliant value minus the uniform-compiler expectation over the region) is zero exactly where compliance leaves no slack and positive exactly where the region is wide and the value varies on it. Registered predictions: singleton regions $\{5\}$ and $\{12\}$ with $U(x) = x$ have dividend 0 with every best-of-k gain 0; the wide-flat task (all odd worlds, U constant 5) has dividend 0 with all gains ≤ 0 closing at 0; the wide-varying tasks have dividend exactly 7 (all even worlds, $U(x) = x$) and exactly 73/11 (popcount ≥ 2 worlds, $U(x) = \text{popcount}(x) * 4 - (x \% 3)$) with weakly increasing gain curves ending exactly at the dividend. gain(1) under the registered ascending scan is deterministic first-element and may be negative (-7 and -15/11 on the wide tasks); the registered claim is about the curve's endpoint and monotonicity, not gain(1) ≥ 0 .

Decision rule: Accept dividend_confirmed if all three gate families hold (DIV_SINGLETON_ZERO, DIV_FLAT_ZERO, DIV_WIDE_POSITIVE). Accept dividend_refuted if any zero-case shows a positive dividend or a wide curve fails to reach its dividend. Otherwise inconclusive. Any recorded verdict passes CI.

Claim boundary: Accepted only as: on the five registered tasks over $\{0,1\}^4$, the choice dividend of D12 is exactly 0 on both singletons and on the flat wide region, exactly 7 and 73/11 on the two varying wide regions, and the registered best-of-k sweep is weakly increasing with endpoint exactly the dividend. This is the kernel arithmetic of P11/D12 with the Theorem-B framing: intelligence pays only on slack. The learner half of Gate 1 - real capability sweeps on real models - is explicitly NOT run here and stays open. No LLM, no valence, no claim about any real delegate.

Recorded signals: choice_dividend_summary.json: gates 6/6 true; choice_dividend_summary.json:ranking.verdict=dividend_confirmed; choice_dividend_summary.json:status=pass

Source: experiments/choice_dividend/preregistration.json

compiler_tomography_pair

Hypothesis: For the concern-parameterised compiler family on the 4-bit Boolean world with hypothesis grid θ in $\{-1,-0.5,0,0.5,1\}^2$ and true $\theta = (0,0)$, (a) MDL over the grid recovers θ with probability ≥ 0.95 by $N=2000$ paired samples averaged over 100 seeds (Theorem CT-1); (b) under the Boltzmann ecology update K_{t+1} proportional to $K_t \exp(\beta r)$ with reward $r(x)=x_0+x_2$ in $\{0,1,2\}$, expected reward per fiber is monotone non-decreasing over 20 steps at every β in $\{0.1, 1.0, 4.0\}$, and converges within 0.05 of the fiber argmax at $\beta=4.0$ (Theorem CT-2).

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: compiler_tomography_pair_summary.json: gates 4/4 true; compiler_tomography_pair_summary.json:status=pass

Source: experiments/compiler_tomography_pair/preregistration.json

concern_fisher_pair

Hypothesis: For the exponential-family concern parameterisation of the 4-bit Boolean world with concern statistic $T(x) = (2x_0 - 1, 2x_2 - 1)$, (a) the empirical Fisher information matrix on every fiber $q^{-1}(z)$ matches the closed-form prediction $g_{\{c,z\}} = \beta^2 \text{diag}(\text{sech}^2(\beta c_1), \text{sech}^2(\beta c_2))$ at every point (c, z) in the pre-registered grid (Theorem CG-1); (b) the non-exact concern-1-form $\alpha + \epsilon$ $(-c_2 \text{ dc}_1)$ has holonomy ϵ along the unit rectangle in c -space and $\epsilon/2$ along the unit triangle, both computed by trapezoidal quadrature on 500 steps per edge (Theorem CG-2 via Green's theorem).

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: concern_fisher_pair_summary.json: gates 4/4 true; concern_fisher_pair_summary.json:status=pass

Source: experiments/concern_fisher_pair/preregistration.json

cross_task_learnability

Hypothesis: For a task family that separates a discrete latent Z on a finite X with min-fibre probability $p_{\min} = 1/(cM)$, the exact recovery probability of empirical common-sufficient clustering is at least $1 - \epsilon$ whenever the sample count N is at least the Theorem-5 bound $\text{ceil}(c M \ln(M / \epsilon))$. Below M samples, recovery is impossible (pigeonhole). The recovery curve is monotone in N .

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: cross_task_learnability_summary.json: gates 4/4 true; cross_task_learnability_summary.json:status=pass

Source: experiments/cross_task_learnability/preregistration.json

cross_task_learnability_continuous

Hypothesis: Theorem 6 (paper section 2.5b) says that continuous-case learnability at resolution ϵ reduces to Theorem 5 via ϵ -covering: sample complexity is $N \geq c N_{\epsilon} \ln(N_{\epsilon} / \epsilon_{\text{rel}})$ with $N_{\epsilon} = O((D_Z / \epsilon)^{d_Z})$, which is polynomial in $1/\epsilon$ at fixed d_Z and exponential in d_Z at fixed ϵ . On an ambient $X = [0,1]^2$ quantised into a 16×16 grid, with a coarser $r \times r$ latent- Z grid (r in $\{4, 8, 16\}$) at intrinsic dimensions d_Z in $\{1, 2\}$: (a) at Theorem 6's bound $N \geq \text{ceil}(c M \ln(M / \epsilon_{\text{rel}}))$ with $c = 1$ and $\epsilon_{\text{rel}} = 0.05$, the exact recovery probability is at least $1 - \epsilon_{\text{rel}}$ for every (d_Z, r) ; (b) recovery is zero below M samples (pigeonhole); (c) recovery is monotone in N up to the bound; (d) at fixed r , $N_{\text{bound}}(d_Z=2, r)$ strictly exceeds $N_{\text{bound}}(d_Z=1, r)$, with the ratio growing at least as $r/2$ (the exponential-in- d_Z scaling).

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: cross_task_learnability_continuous_summary.json: gates 5/5 true; cross_task_learnability_continuous_summary.json:status=pass

Source: experiments/cross_task_learnability_continuous/preregistration.json

cross_task_sufficiency

Hypothesis: On a 4-bit Boolean world with latent $Z = (\text{parity}\{0,1\}, \text{parity}\{2,3\})$, a task family whose members all factor through Z admits Z itself as the coarsest common sufficient statistic; a task family whose members instead reveal individual bits admits only the identity as a common sufficient statistic; and for the shared family the coarsest common sufficient statistic is strictly finer than the minimal sufficient statistic of any single task in the family.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: cross_task_sufficiency_summary.json: gates 4/4 true; cross_task_sufficiency_summary.json:status=pass

Source: experiments/cross_task_sufficiency/preregistration.json

delete_repair_concern

Hypothesis: Theorem 4 is silent between bag's five representing screens; Paper F broke the tie by name and door 1 showed that tie-break is not relabel-natural. A registered concern weighting gives the choice a reason: distinct concerns select distinct screens, the concern-free choice is strictly costly somewhere, the mirrored pair is reversal-natural, and the bag/pair_eq dial has an exact threshold at 11/27. This opens the close-out's third job (care which matter) at the lowest bound, as weights and nothing more.

Decision rule: Accept concern_does_work if at least three distinct screens are selected across the six registered concerns, the mirrored pair is reversal-natural, and the unweighted choice is strictly beaten with an exact recorded gap. Accept concern_idle if every concern selects one screen. Otherwise inconclusive. Any recorded verdict passes CI.

Claim boundary: Accepted only as: on this menu with this registered cost, the screen worth holding is a function of the registered task weighting; the concern-free tie-break is strictly costly under registered concerns; the choice layer is reversal-natural where the name tie-break was not. Concern is a weight vector. Not valence, not agency, not consciousness, not learned, not an LLM, not a new master object, not Paper G.

Recorded signals: delete_repair_concern_summary.json: gates 9/9 true; delete_repair_concern_summary.json:ranking.verdict=concern_does_work; delete_repair_concern_summary.json:status=pass

Source: experiments/delete_repair_concern/preregistration.json

delete_repair_concern_estimation

Hypothesis: Door 3 opened the third job with concern as a given weight vector and licensed exactly one next step: a registered instrument for estimated concern. This is that instrument. Concern is estimated by frequency counting on three registered literal 24-draw sequences; the plug-in choice is κ concern of the empirical frequencies on door 3's unchanged menu and cost rule. Registered predictions: the plug-in choice equals the oracle choice for all prefixes $n \geq$ a minimal step of 1 (SEQ_BAG), 2 (SEQ_MIX), and 6 (SEQ_PAIR, where the odd-prefix pair_eq frequency $k/(2k+1)$ crosses the exact 11/27 boundary between $n = 5$ and $n = 7$); and holding SEQ_MIX's screen q_{stab0} under SEQ_PAIR's true concern costs exactly 4 more than q_{id} (20 vs 16).

Decision rule: Accept estimation_works if all three sequences converge to the oracle choice at exactly the registered minimal steps (1, 2, 6) and stay through $n = 24$. Accept estimation_fails if any final-prefix choice differs from oracle. Otherwise inconclusive. Any recorded verdict passes CI.

Claim boundary: Accepted only as: on door 3's menu and cost rule, with these three registered draw sequences, registered frequency counting recovers the oracle concern choice at exactly the registered minimal prefixes (1, 2, 6) and holding the wrong sequence's screen costs exactly 4. Frequency counting is the entire estimator. Not SGD, not valence, not learned representations, not an LLM, not a new master object, not Paper G.

Recorded signals: delete_repair_concern_estimation_summary.json: gates 8/8 true; delete_repair_concern_estimation_summary.json:ranking.verdict=estimation_works; delete_repair_concern_estimation_summary.json:status=pass

Source: experiments/delete_repair_concern_estimation/preregistration.json

delete_repair_concern_transport

Hypothesis: The concern choice function's symmetry layer is coordinate-free (reversal-natural under both menus) while its content layer is menu-relative: the bag/pair_eq boundary moves from 11/27 (to q_{id}) on the base menu to 7/27 (to q_{pair01}) on the extended menu, because q_{pair01} serves pair_eq at cost 12 instead of the 32 miss penalty. Doors 1 and 3 compose: what to hold is a joint function of the menu and the expected questions.

Decision rule: Accept transport_holds_boundary_moves if reversal naturality holds under both menus, both registered boundaries are confirmed by exact sweeps, and the boundaries differ. Accept boundary_menu_stable if naturality holds and they are equal. Accept transport_fails if naturality breaks. Otherwise inconclusive. Any recorded verdict passes CI.

Claim boundary: Accepted only as: the concern choice's symmetry layer transports across the reversal under both registered menus, its phase boundary is menu-relative (11/27 vs 7/27, to different screens), and the concern-free anchor is menu-stable. Not valence, not learned concern, not an LLM, not a new master object, not Paper G.

Recorded signals: delete_repair_concern_transport_summary.json: gates 7/7 true; delete_repair_concern_transport_summary.json:ranking.verdict=transport_holds_boundary_moves; delete_repair_concern_transport_summary.json:status=pass

Source: experiments/delete_repair_concern_transport/preregistration.json

delete_repair_connection

Hypothesis: Aff(1, Z/3) path-ordered transport is not integer Kirchhoff. Additive cycles remain the Paper A control. Affine cycles that escape the sum-b prediction make cell 3 real on this harness.

Decision rule: Accept cell3_holds only if the Kirchhoff control holds, at least one affine cycle escapes it, composition does not commute, raw comparison fails on a flat affine section, and transported comparison works. Otherwise cell3_idle. A kill is a valid passing run.

Claim boundary: Accepted only as: cell 3 is not idle Kirchhoff packaging on this Aff(1, Z/3) 4-cycle. Not Lorentz. Not CG-2. Not Paper D/E/F. Not a better LLM.

Recorded signals: delete_repair_connection_summary.json: gates 5/5 true; delete_repair_connection_summary.json:ranking.verdict=cell3_holds; delete_repair_connection_summary.json:status=pass

Source: experiments/delete_repair_connection/preregistration.json

delete_repair_disanalogy

Hypothesis: The shared cartoon is not a shared theorem. On the diamond fibre, the causal poset does not fix $s(e_1, e_2)$. Quotienting positions still fails first_bit. Unique s , or a working PE quotient, reopens Possibility 6.

Decision rule: Accept disanalogy_holds only if the diamond fibre has at least two s values, concurrency of $\{e_1, e_2\}$ is constant, and q_{perm} fails first_bit. Otherwise identification_reopened. A reopen is a valid passing run.

Claim boundary: Accepted only as: the shared cartoon is not a shared theorem on this harness. Not a functor. Not real Lorentz physics. Not Paper E/F. Not a better LLM.

Recorded signals: delete_repair_disanalogy_summary.json: gates 5/5 true; delete_repair_disanalogy_summary.json:ranking.verdict=disanalogy_holds; delete_repair_disanalogy_summary.json:status=pass

Source: experiments/delete_repair_disanalogy/preregistration.json

delete_repair_generators

Hypothesis: Generator episodes form a border, not a one-off: the cube macro shows the same two-ledger shape as the banked sq episode - screens invariant on the shared universe, access changed, round trip identity - with min sizes predicted in advance (5 vs 2 for x^3 at bound 5; the sq replay must give 7 vs 3 for x^4 at bound 7).

Decision rule: Accept border_consolidated if both episodes are outside facts. Accept border_sharpened if exactly one reduces to screen motion. Otherwise inconclusive. Any recorded verdict passes CI.

Claim boundary: Accepted only as: two generator episodes, same border - access facts move with the generator set while all registered (q, K) data is invariant. Not a kill of the representability reading. Not continuum. Not an LLM. Not Paper G.

Recorded signals: delete_repair_generators_summary.json: gates 7/7 true; delete_repair_generators_summary.json:ranking.verdict=border_consolidated; delete_repair_generators_summary.json:status=pass

Source: experiments/delete_repair_generators/preregistration.json

delete_repair_kappa

Hypothesis: A cheap five-field signature is not a function to gold. A menu-relative coarsest-representing map is a function, is natural under 03 relabelling, and is Theorem 4 plus a total order. Uniqueness without that tie-break fails. Possibility 1 as a new master object dies; the written function is SIC.

Decision rule: Accept calculus_is_sic if kappa_cheap is not a function, kappa_screen matches gold on every registered case, uniqueness fails without the tie-break, and Path A/B disagree. Accept calculus_holds if the cheap signature is already a function. Accept no_function if kappa_screen misses. Any of those three is a valid passing run.

Claim boundary: Accepted only as: the written function is SIC's screen plus a total order. Not a new master object. Not text nomination. Not an LLM. Not a better language model.

Recorded signals: delete_repair_kappa_summary.json: gates 6/6 true; delete_repair_kappa_summary.json:ranking.verdict=calculus_is_sic; delete_repair_kappa_summary.json:status=pass

Source: experiments/delete_repair_kappa/preregistration.json

delete_repair_menu_blind

Hypothesis: Gold is menu-relative. On a fixed (task, screen) case, enlarging the menu flips gold from noop to quotient. Any menu-blind kappa is a function of (Y, q) alone and is therefore constant across menus, so it cannot match gold on both. This closes the close-out's door 1 categorically rather than at five fields, without refitting the cheap rule.

Decision rule: Accept menu_blind_dead if at least one gold flip exists between MENU_BASE and MENU_EXT and kappa_screen (menu-relative) is exact on all 34 rows. Accept menu_blind_lives if no flip exists and the frozen five-field rule is exact and collision-free under both menus. Accept no_function if kappa_screen misses a row. Otherwise inconclusive. Any recorded verdict passes CI.

Claim boundary: Accepted only as: gold is menu-relative on this harness, so no menu-blind kappa of any signature width is correct across menus; the five-field rule's collision and correctness are menu-relative; kappa_screen stays exact per menu; the name tie-break is not relabel-natural on ties. Not a new master object. Not a refit. Not an LLM. Not Paper G.

Recorded signals: delete_repair_menu_blind_summary.json: gates 11/11 true; delete_repair_menu_blind_summary.json:ranking.verdict=menu_blind_dead; delete_repair_menu_blind_summary.json:status=pass

Source: experiments/delete_repair_menu_blind/preregistration.json

delete_repair_reduction

Hypothesis: There is a delete-repair fact outside (q, K) at this bound: the squaring-macro episode changes min formula size and fibre mass while every registered screen partition on the shared universe is unchanged and the round trip is the identity. The fact lives on the generator set - the process coordinate the close-out already banked as process-relative - so Possibility 5's dynamics reading is bounded, not killed: representability facts move (q, K); generator-episode access facts do not.

Decision rule: Accept outside_fact_found if the shared universe embeds, all three registered screens are extensionally identical across grammars on it, the delete/re-adjoin round trip is the identity on the access table, and min size or mass differs between grammars. Accept all_reduce if a registered screen moves and the access change tracks it. Otherwise inconclusive. Any recorded verdict passes CI.

Claim boundary: Accepted only as: at this bound, the squaring episode is a delete-repair fact whose observable is invariant under all registered (q, K) motions and changes with the generator set. Possibility 5's dynamics reading is thereby bounded to representability facts, exactly where the close-out's process-relative split already put the border. Not a kill of the representability reading. Not continuum. Not an LLM. Not Paper G.

Recorded signals: delete_repair_reduction_summary.json: gates 8/8 true; delete_repair_reduction_summary.json:ranking.verdict=outside_fact_found; delete_repair_reduction_summary.json:status=pass

Source: experiments/delete_repair_reduction/preregistration.json

delete_repair_surgery

Hypothesis: A name-blind one-shot rule compiled from the three-cell taxonomy is not a complete agent decision procedure. Empirical gold is menu-relative representability. Unused symmetry (pair_eq on q_id) is leftover-looking and not quotientable on the registered menu. If every held-out row matches, surgery_holds. A miss is surgery_killed. Either verdict passes CI.

Decision rule: Accept surgery_holds only if the signature rule matches empirical gold on every held-out row. Construction rows are disclosed authoring toys and cannot save a miss. A surgery_killed verdict is a valid passing run.

Claim boundary: Accepted only as: on this harness, the cheap taxonomy signature is not a complete one-shot agent rule. Not text nomination. Not an LLM result. Not Paper F. Not a better LLM. Paper B is not reopened.

Recorded signals: delete_repair_surgery_summary.json: gates 7/7 true; delete_repair_surgery_summary.json:ranking.verdict=surgery_killed; delete_repair_surgery_summary.json:status=pass

Source: experiments/delete_repair_surgery/preregistration.json

delete_repair_swap

Hypothesis: Typed restore-distinction and typed quotient-privilege are not interchangeable on the Paper A harness. Crossed success, or one minimal-safe screen for both toys, kills the taxonomy as a discriminator.

Decision rule: Accept taxonomy_holds only if typed repairs work, the crossed over-repair fails representability, the under-quotient is strictly cheaper, and no single screen is minimal-safe for both. Otherwise taxonomy_killed. A kill is a valid passing run.

Claim boundary: Accepted only as the Paper B swap-cell discriminator on this harness. Not new enumeration. Not a universal calculus. Not text nomination. Not Paper C/D/E/F. Not Lorentz = Lamport = positional encodings.

Recorded signals: delete_repair_swap_summary.json: gates 6/6 true; delete_repair_swap_summary.json:ranking.verdict=taxonomy_holds; delete_repair_swap_summary.json:status=pass

Source: experiments/delete_repair_swap/preregistration.json

delete_the_absolute

Hypothesis: On $X = \{0,1\}^4$ with orbit-canonical screens q_G , exact representability of a task Y occurs iff model symmetry is contained in target symmetry ($G_M \subseteq G_Y$). Under-repair ($G_M \not\subseteq G_Y$) makes Y unrepresentable: Y is not constant on q -fibres. Over-repair (G_M a proper subset of G_Y) remains representable but has strictly more fibres (leftover privilege / shortcut surface). Exact repair of a mixed fibre must split leftover Y -disagreement. Delete-then-default and relative-then-drop do not commute. A 4-cycle of integer steps closes iff the steps sum to 0, and two potentials for the same steps differ by a constant.

Decision rule: Use registered gates

Claim boundary: Accepted only as a finite exact regression of the listed DeleteRepair facts on $n=4$ binary sequences. Not a general proof for arbitrary n or groups. Not Shannon entropy. Not a relativity theorem. Over-repair is leftover privilege, not an obstruction.

Recorded signals: delete_the_absolute_summary.json: gates 7/7 true; delete_the_absolute_summary.json:status=pass

Source: experiments/delete_the_absolute/preregistration.json

dial_nestedness

Hypothesis: Review item 2 on 'Intention Is All You Need' v3: Theorem B's 'as D grows the cells coarsen' implies nested optima that rate-distortion does not guarantee. Registered predictions: rates (5, 3, 2, 2, 1) on the budget grid (0, 1/4, 1/2, 3/4, 1); $D=0$ optimum uniquely the level-set partition; all-optimizer nesting FAILS (registered witness shape: a $\{01\}_{23}\{4\}$ -type optimizer at 1/4 does not refine a $\{012\}_{34}$ -type optimizer at 1/2); a chosen chain of optimizers exists that does nest. The exported wording is only: the optimal rate falls.

Decision rule: Accept nestedness_fails_generally if the rate vector is weakly decreasing and some adjacent optimizer pair fails refinement. Accept nestedness_holds_here if every adjacent optimizer pair nests. Otherwise inconclusive. chain_exists is recorded separately in every case. Any recorded verdict passes CI.

Claim boundary: Accepted only as: on this registered world, the optimal rate falls (5, 3, 2, 2, 1), the $D=0$ optimum is the level partition, all-optimizer nesting fails with an explicit witness, and a nested chain exists as a selection. The essay's 'cells coarsen' wording stays withdrawn; the exported general slogan is 'the optimal description rate falls as the budget grows'. Not a continuous claim. Not average-case. Not an LLM.

Recorded signals: dial_nestedness_summary.json: gates 8/8 true; dial_nestedness_summary.json:ranking.verdict=nestedness_fails_generally; dial_nestedness_summary.json:status=pass

Source: experiments/dial_nestedness/preregistration.json

eml_fiber_spectrum

Hypothesis: The homogeneous EML grammar $S \rightarrow 1 \mid \text{eml}(S,S)$ has no 1-D integer invariant like polynomial degree. Exhaustive enumeration through 6 internal nodes matches Catalan counts. Two equal-size trees disagree at a registered closed evaluation. Size is therefore not a denotation invariant. US-4 is not tested.

Decision rule: Accept the constant-grammar census if and only if Catalan enumeration matches and a same-size different-value witness exists. Do not promote US-4. Cross-size collision is recorded if a well-resolved exact identity is observed and is not a fatal gate.

Claim boundary: Accepted only as a constant-grammar census through $k=6$ plus an exact same-size disagreement. Not a variable- x spectrum. Not US-4. Not an identity theorem for rounded collisions. Not a 1-D invariant.

Recorded signals: eml_fiber_spectrum_summary.json: gates 4/4 true;
eml_fiber_spectrum_summary.json:optional_gates.EFS_CROSS_SIZE_COLLISION.status=pass; eml_fiber_spectrum_summary.json:status=pass

Source: experiments/eml_fiber_spectrum/preregistration.json

eml_us4_discrete

Hypothesis: If unknown-skeleton GD tied because 1-leaves were free, greedy frozen-leaf rewrite should reopen a Phi gap on extra basins. Equal extra counts reject that transfer at this bound.

Decision rule: Accept phi_holds only if zero_extra > thin_extra. Equal extras reject (min_size_governs). thin_extra > zero_extra kills. Exact 2-vs-1 is the control and cannot decide the claim.

Claim boundary: Accepted only as the preregistered extra-basin ranking of frozen-leaf greedy rewrite. Not a Gibbs tautology. Not unknown-skeleton GD. Not the neural bootstrap. Not a general architecture law.

Recorded signals: eml_us4_discrete_summary.json: gates 6/6 true; eml_us4_discrete_summary.json:ranking.verdict=phi_holds;
eml_us4_discrete_summary.json:status=pass

Source: experiments/eml_us4_discrete/preregistration.json

eml_us4_gradient

Hypothesis: On the variable- x EML language, two min-internal=3 targets with different truncated Gibbs mass Phi have different master-formula gradient-recovery success if fiber mass governs access. Equal recovery means min-size still governs and Phi-predicts-GD dies at this bound. Singleton beating zero kills Phi-predicts-GD. This is a local-CPU analogue, not Odrzywoek's neural bootstrap, and not a Gibbs sampler of the census.

Decision rule: Apply the ranking rule below after the run, never before. Do not retune seeds, inits, steps, or the success threshold after seeing counts. If perturbed-correct fails, withhold (do not reject). If blind counts are equal, reject Phi-predicts-GD and record min-size-still-governs. Do not dress a null as a win.

Claim boundary: At most a local-CPU matching-skeleton GD ranking of two registered size-3 targets. Not Odrzywoek's neural bootstrap. Not a Gibbs-sampler result. Not function identity from the grid except the exact zero identity. Not a claim that Phi governs every optimizer.

Recorded signals: eml_us4_gradient_summary.json: gates 6/6 true; eml_us4_gradient_summary.json:ranking.verdict=phi_holds;
eml_us4_gradient_summary.json:status=pass

Source: experiments/eml_us4_gradient/preregistration.json

eml_us4_prime

Hypothesis: On the truncated variable- x EML language ($k \leq 5$, $\pi(t)4^{\{-|t\}}$), Gibbs access $P(z)=\text{Phi}_z/Z$ is not a function of shortest depth. Same-min-size finite fibers can differ in mass by a factor of at least 2 because of min-shell multiplicity. Extra shells and master-formula gradient recovery are not claimed.

Decision rule: Accept the EML-native Gibbs-vs-shortest split if and only if the registered $k=3$ finite class has $\text{Phi}_{\max}/\text{Phi}_{\min} \geq 2$ and the zero identity holds. Do not promote gradient recovery. Do not claim the Sq extra-shell mechanism transferred.

Claim boundary: Accepted only as: on this truncated language, Gibbs mass is not a function of shortest depth, and the zero function has two size-3 formulas. Not gradient recovery. Not the Sq extra-shell transfer. Not function identity from the grid except the exact zero identity.

Recorded signals: eml_us4_prime_summary.json: gates 6/6 true; eml_us4_prime_summary.json:status=pass

Source: experiments/eml_us4_prime/preregistration.json

eml_us4_search

Hypothesis: If access is fiber mass, blind GD over every size-3 skeleton should recover the fat zero target from more skeletons than the thin singleton. Equal skeleton counts reject that transfer at this bound.

Decision rule: Accept phi_holds only if perturbed-correct succeeds and zero_gd > thin_gd. Equal counts reject Phi-predicts-unknown-skeleton-GD (min_size_governs). thin > zero kills it. Optimizer failure withholds.

Claim boundary: Accepted only as the preregistered ranking of unknown-skeleton GD. Not a Gibbs tautology. Not matching-skeleton GD. Not the neural bootstrap. Not a general architecture law.

Recorded signals: eml_us4_search_summary.json: gates 7/7 true; eml_us4_search_summary.json:ranking.verdict=min_size_governs;
eml_us4_search_summary.json:status=pass

Source: experiments/eml_us4_search/preregistration.json

eml_variable_spectrum

Hypothesis: The leaf-labeled EML grammar $S \rightarrow 1 \mid x \mid \text{eml}(S, S)$ has no 1-D integer invariant like degree. Exhaustive enumeration through 5 internal nodes matches $2^{\lfloor k+1 \rfloor} C_k$. Two equal-size trees disagree as functions on a registered positive grid. US-4 is not tested.

Decision rule: Accept the variable- x census if and only if labeled Catalan counts match and the registered size-1 pair disagrees at $x=2$. Do not promote US-4. Finite-grid agreement is not function identity.

Claim boundary: Accepted only as a leaf-labeled census through $k=5$ plus an exact same-size functional disagreement. Not US-4. Not identity of functions. Not a 1-D invariant. Not the constant-only grammar.

Recorded signals: eml_variable_spectrum_summary.json: gates 5/5 true; eml_variable_spectrum_summary.json:status=pass

Source: experiments/eml_variable_spectrum/preregistration.json

interventional_crl_learnability

Hypothesis: SIC-C-c asks whether restricting to a specific inductive-bias hypothesis class can restore uniform polynomial-in- d_Z sample complexity for the continuous-case learnability problem of paper section 2.5b (Theorem 6). Instruments 8, 9, and 10 answered yes for the linear-ICA, sparse-ICA / IMA, and iVAE classes. This instrument (interventional CRL, Ahuja--Mahajan--Wang--Bengio 2022; Squires et al. 2023) does the same for the interventional-causal-representation-learning class. Setup: latent Z with independent $\text{Laplace}(0, 1)$ marginals under environment $E = 0$ (observational); for each i in $\{1, \dots, d_Z\}$, environment $E = i$ intervenes on Z_i by drawing it from $\text{Laplace}(\mu_i, 0.5)$ with the other Z_j ($j \neq i$) still $\text{Laplace}(0, 1)$, where μ_i is drawn once per (seed, d_Z) uniformly on $[-2, 2]$. Nonlinear mixing $X = W_2 \tanh(\alpha W_1 Z)$ with $\alpha = 0.5$ and W_1, W_2 Haar-orthogonal via signed QR. Algorithm (simplified interventional CRL): fit sklearn FastICA per environment; align each intervention environment's unmixing by the mean-shift-max row heuristic (the row whose distribution shifts most between $E = 0$ and $E = i$ is identified as recovering Z_i), stack d_Z aligned rows into a $d_Z \times d_Z$ environment-consistency unmixing; score by $\text{Amari}(W_{\text{aligned}} @ A_{\text{linear}})$ with $A_{\text{linear}} = W_2 @ W_1$ (the Jacobian of the MLP at $Z = 0$). Pooled control: fit FastICA on pooled X ignoring E ; score by $\text{Amari}(W_{\text{pool}} @ A_{\text{linear}})$. Predictions: (a) environment-split Amari ≤ 0.20 at $N_{\text{per_env}} = 10000$ for every d_Z in $\{2, 3, 4\}$; (b) at $N_{\text{per_env}} = 10000$ the environment-split Amari is strictly less (by more than 0.005) than the pooled control for every d_Z ; (c) at every d_Z the environment-split Amari($N_{\text{per_env}}$) curve is nonincreasing modulo a 0.03 wobble; (d) $\log(N_{\text{per_env_needed}}) = a + b \log(d_Z)$ fitted at target Amari 0.30 has slope $b \leq 5.0$.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: interventional_crl_learnability_summary.json: gates 4/4 true; interventional_crl_learnability_summary.json:status=pass

Source: experiments/interventional_crl_learnability/preregistration.json

ivae_learnability

Hypothesis: SIC-C-c asks whether restricting to a specific inductive-bias hypothesis class can restore uniform polynomial-in- d_Z sample complexity for the continuous-case learnability problem of paper section 2.5b (Theorem 6). Instrument 8 answered yes for linear ICA and instrument 9 answered yes for the sparse-mixing / independent-mechanism-analysis class; this instrument (instrument 10) answers the same for the auxiliary-variable identifiable-ICA / iVAE class of Khemakhem, Kingma, Monti, Hyvarinen 2020. Setup (simplified auxiliary-variable identifiable model): latent Z in $\mathbb{R}^{d_Z}$ with $Z_i \mid U \sim \text{Laplace}(\mu_i(U), 1)$, auxiliary variable U discrete in $\{0, \dots, K-1\}$ with $K = 2 d_Z$, per- u offsets $\mu_i(u)$ drawn from $\text{Uniform}([-3, 3])^{d_Z}$ once per (d_Z , trial), and linear mixing $X = A Z$ with A Haar-orthogonal on $d_Z \times d_Z$. Algorithm: per-conditional linear-ICA fallback. For each u we run sklearn FastICA(parallel, logcosh, whiten='unit-variance', max_iter=2000, tol=1e-6, random_state=trial1000+u) on the samples $X \mid U = u$; buckets with fewer than $\max(2d_Z, 10)$ samples are skipped and the count is reported per grid cell. The global metric averages $\text{Amari}(W_u A)$ across the non-skipped u values (the Amari index is invariant to signed permutation, which absorbs the per- U rotation ambiguity of linear ICA). Predictions: (a) averaged Amari ≤ 0.10 at $N = 10000$ for every d_Z in $\{2, 4, 6\}$; (b) for every d_Z the Amari(N) curve is nonincreasing modulo a 0.02 wobble tolerance; (c) the smallest N in the sweep $\{200, 500, 1000, 2000, 5000, 10000\}$ reaching averaged Amari ≤ 0.15 grows only polynomially in d_Z (fitted log-log slope $b \leq 4.0$); (d) at $d_Z = 6$ the iVAE class sample complexity to reach Amari ≤ 0.15 is at least 5x below the Theorem 6 exponential bound $\text{ceil}((D_Z / \text{eps})^{d_Z} \ln((D_Z / \text{eps})^{d_Z} / \text{eps_rel}))$ with $D_Z = 1$, $\text{eps} = 0.25$, $\text{eps_rel} = 0.05$, $c = 1$ ($= 46341$ samples). Gates are looser than instruments 8/9 because per-conditional splitting at $K = 2 * d_Z$ reduces the effective sample size per ICA fit; the honest thresholds match the realism budget spelled out in the prompt.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: ivae_learnability_summary.json: gates 4/4 true; ivae_learnability_summary.json:status=pass

Source: experiments/ivae_learnability/preregistration.json

linear_ica_learnability

Hypothesis: SIC-C-c asks whether restricting to a specific inductive-bias hypothesis class can restore uniform polynomial-in- d_Z sample complexity for the continuous-case learnability problem of paper section 2.5b (Theorem 6). For the linear-ICA class --- latent Z with independent non-Gaussian marginals ($\text{Laplace}(0, 1)$), square orthogonal mixing matrix A , observation $X = A Z$ --- sklearn FastICA on N samples of X , evaluated by the Amari index $\text{amari}(P)$ with $P = W A$, is predicted to (a) achieve Amari ≤ 0.02 at $N = 10000$ for every d_Z in $\{2, 4, 6, 8\}$; (b) reach the polynomial-fit threshold Amari ≤ 0.03 with a smallest- N that grows only polynomially in d_Z (fitted log-log slope $b \leq 3.0$); (c) yield a nonincreasing Amari(N) curve at every d_Z modulo a 0.01 numerical wobble; and (d) at $d_Z = 8$ land the linear-ICA sample complexity at least 20x below the Theorem 6 exponential bound $\text{ceil}((D_Z / \text{eps})^{d_Z} * \ln((D_Z / \text{eps})^{d_Z} / \text{eps_rel}))$ with $D_Z = 1$, $\text{eps} = 0.25$, $\text{eps_rel} = 0.05$, $c = 1$.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: linear_ica_learnability_summary.json: gates 4/4 true; linear_ica_learnability_summary.json:status=pass

Source: experiments/linear_ica_learnability/preregistration.json

pac_bayes_weakness_enum

Hypothesis: On the four frozen finite domains with $|X|=|Y| \leq 7$, exact enumeration of $H=Y^X$ recovers the overlapping-mixture mass $P(h) = \sum_{k \leq W(h)} \pi_k / |H_{\geq k}|$, the cyclic identity-rho fixed-action class has size $|Y|^r = 7$, and the Langford-Seeger-Maurer plug-in at $m=8$, $\delta=0.05$ is non-vacuous for the aligned truth. Wrong/random groups, pi-schedule sign flips, and a hyperprior that merely repeats W_{aligned} are registered as independent kills of the OOD / extra-complexity readings, not of the finite mass formula.

Decision rule: instrument_failed if class counts, the mass formula, or the cyclic fixed-action count break. finite_iid_killed if any $m=8$ aligned-truth bound is ≥ 0.99 . finite_iid_holds_ood_or_weight_killed if the IID lane is non-vacuous but a wrong/random certificate is at least as tight as the aligned truth while the shortcut fails OOD, or if pi flips the truth-vs-shortcut KL sign. finite_iid_holds otherwise. hyperprior_no_extra_info and neural_untransported are recorded independently. Any completed honest verdict other than a broken instrument passes CI.

Claim boundary: Accepted only as a finite exact enumeration of the sketch's mixture-mass certificate and the registered IID/OOD/weight/hyperprior kills. Not a proof of Langford-Seeger-Maurer. Not a neural PAC-Bayes bound. Not a relabel of the existing 500-trial symbolic OOD tables. Analytic-open P0 leftovers stay open.

Recorded signals: pac_bayes_weakness_enum_summary.json: gates 9/10 true;
pac_bayes_weakness_enum_summary.json:ranking.verdict=finite_iid_holds_ood_or_weight_killed;
pac_bayes_weakness_enum_summary.json:status=pass

Source: experiments/pac_bayes_weakness_enum/preregistration.json

rate_distortion_pair

Hypothesis: Theorem 2 (paper section 2.2) says the Shannon rate-distortion pair (q_D, K_D) is a stochastic fibration for every distortion budget $D \geq 0$ -- a one-parameter deformation of the sufficiency fibration Theorem 1 provides, with q_0 minimal-sufficient and $q_{D_{\max}}$ the constant map. For two finite sources (uniform on $n=4$ symbols; Bernoulli($p=0.3$)) under Hamming distortion, the RD function $R(D)$ has an exact closed form; the RD-optimal test channel achieves $I(X; \hat{X}) = R(D)$ at every distortion budget in the achievable regime; $R(0)$ equals the source entropy (Theorem 1 anchor); $R(D_{\max}) = 0$ (encoder collapses to a constant); R is monotone nonincreasing in D ; and R is convex on the grid.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: rate_distortion_pair_summary.json: gates 10/10 true; rate_distortion_pair_summary.json:status=pass

Source: experiments/rate_distortion_pair/preregistration.json

representation_repair_pair

Hypothesis: For each of the eight canonical (failure_signature, minimal_lift) pairs listed in Extended-Program section 5.8 of The Structural Intelligence Conjecture (scalar->operator, global norm->localized measure, quotient->restored fiber, static->path space, affine->projective, point->ensemble, non-composing->interface, symmetry->gauge-fix), there is a hand-designed finite toy world on which (a) the broken representation R misses the target invariant I (exists $s \models s'$ with $R(s)=R(s')$ and $I(s) \neq I(s')$), (b) the lifted representation R' captures I (for every $s, s', R'(s)=R'(s')$ implies $I(s)=I(s')$), and (c) R' is minimal in the sense that no strictly smaller enlargement of R also captures I (Theorem RR-1). Additionally, on the product of the scalar->operator world and the static->path-space world, the composition of their independent lifts captures the joint invariant, is itself minimal, and commutes across application orderings (Theorem RR-2).

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: representation_repair_pair_summary.json: gates 4/4 true; representation_repair_pair_summary.json:status=pass

Source: experiments/representation_repair_pair/preregistration.json

representation_search

Hypothesis: Over a Boolean world with a known invariant, a selector that keeps task information ($H(Y|q(X))=0$) and then minimizes description length recovers the ground-truth quotient, whereas pure description-length minimization collapses the obstruction and pure mutual-information maximization never compresses.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: representation_search_summary.json: gates 4/4 true; representation_search_summary.json:status=pass

Source: experiments/representation_search/preregistration.json

sica_finite_derivation_pair

Hypothesis: On the 4-bit Boolean world ($|X| = 16$) with parameter set $\Theta = \{(0,0), (0,1), (1,0), (1,1)\}$ of joint-parity concentrations and Laplace-smoothed pmf $P(\theta, x) = (P_{\hat{\theta}}(\theta, x) + 1/16) / 2$ where $P_{\hat{\theta}}$ concentrates mass $1/4$ on the 4 worlds sharing θ 's joint parity, the LR-vector against $\theta_0 = (0, 0)$ together with the uniform-on-fibre kernel jointly realise SIC-A: (a) T1's cross-multiplication characterisation is a biconditional on all 256 world pairs; (b) K's fibre support is a biconditional on all $4 * 16 = 64$ (z, x) pairs; (c) K sums to 1 on every fibre; (d) the LR-vector-induced fibre partition equals the joint-parity reference MSS partition ($\text{parity}\{0,1\}(x), \text{parity}\{2,3\}(x)$) bit-exact -- four fibres of four worlds each.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: sica_finite_derivation_pair_summary.json: gates 4/4 true; sica_finite_derivation_pair_summary.json:status=pass

Source: experiments/sica_finite_derivation_pair/preregistration.json

sicc_covering_meta_pair

Hypothesis: The SIC-C-c covering meta-theorem (StructuralIntelligenceMathlib.sicc_covering_meta) predicts $n = c K \log(K / \delta)$ as the sample complexity to recover the fibration on an eps-cover of size K. On the coupon-collector realisation of eps-cover recovery in a K-cell rotational latent partition of a 2-D continuous space, the fitted constant $c_{\text{fitted}}(K) = n_{\text{emp}}(K, \delta) / (K * \log(K / \delta))$ is stable to within 20% across K in {8, 16, 32, 64, 128, 256} with $\delta = 0.05$, and each n_{emp} meets exact recovery probability $\geq 1 - \delta$ while $(n_{\text{emp}} - 1)$ does not (the empirical sample count is tight).

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: sicc_covering_meta_pair_summary.json: gates 4/4 true; sicc_covering_meta_pair_summary.json:status=pass

Source: experiments/sicc_covering_meta_pair/preregistration.json

silent_substitution

Hypothesis: Gate 2 of the essay 'Intention Is All You Need' (the silence gate) holds at the kernel in the zero-leakage limit: on a realization space of eight compliant outcomes forming a single compliance class, the spec-level record is the constant symbol 'compliant' by construction, so a registered reward tilt $w_t(x) = (r(x) + 1) ** t$ ($t = 0..12$) drives the misaligned arm's expected delegate reward $E_t[R_{\text{MIS}}]$ strictly up and the expected principal value $E_t[U]$ strictly down ($7/2$ at $t = 0$ to $1583088700/7083249971$ at $t = 12$) while every report stays green; the mass on the reward argmax $x = 7$ strictly rises to exactly $17179869184/21249749913$ at $t = 12$ (above the registered floor $4/5$; the mass tends to 1 in t and first crosses $99/100$ at $t = 35$, outside this horizon); and the aligned control ($R_{\text{AL}} = U$) sends $E_t[U]$ strictly up through the same channel, showing the channel, not the tilt, sets the direction.

Decision rule: Accept substitution_silent if the record is constant across all 13 steps of both arms AND misaligned $E_t[U]$ strictly falls while $E_t[r]$ strictly rises AND the aligned control's $E_t[U]$ strictly rises. Accept substitution_visible if the record ever varies. Otherwise inconclusive. Any recorded verdict passes CI.

Claim boundary: Accepted only as: at the zero-leakage limit - the whole space one compliance class, the record constant by construction - the registered reward tilt drives misaligned expected reward strictly up and expected principal value strictly down while every report stays green, the mass concentrates on the reward argmax exactly as registered, and the aligned control rises through the same channel. This banks the ZERO-LEAKAGE limit only: measuring real specifications' leakage is the open empirical half of Gate 2. No LLM, no valence, no claim about any real delegate; the bridge from record silence to intention attribution stays a bet per the essay 'Intention Is All You Need'.

Recorded signals: silent_substitution_summary.json: gates 8/8 true; silent_substitution_summary.json:ranking.verdict=substitution_silent; silent_substitution_summary.json:status=pass

Source: experiments/silent_substitution/preregistration.json

sparse_ica_learnability

Hypothesis: SIC-C-c asks whether restricting to a specific inductive-bias hypothesis class can restore uniform polynomial-in- d_Z sample complexity for the continuous-case learnability problem of paper section 2.5b (Theorem 6). Instrument 8 answered yes for the linear-ICA class; this instrument (instrument 9) answers the same for the sparse-mixing / independent-mechanism analysis (IMA) class (Gresele et al. 2021). Setup: latent Z with independent Laplace(0, 1) marginals; mixing A drawn as (i) a Haar-orthogonal Q via signed QR, (ii) multiplied entry-wise by a Bernoulli(s) sparse mask with sparsity s in {0.5, 0.25} and rank-deficiency retries (or a keep-diagonal fallback), (iii) re-orthonormalised via a second signed QR; observation $X = A Z$. Algorithm: *sklearn FastICA* (*parallel, logcosh, whiten='unit-variance', max_iter=2000, tol=1e-6, random_state=trial*). Metric: Amari index on $P = W A$. Predictions: (a) averaged Amari ≤ 0.02 at $N = 10000$ for every d_Z in {2, 4, 6, 8} and every sparsity in {0.5, 0.25}; (b) sparser (s=0.25) recovery is at least as good as denser (s=0.5) modulo a 0.01 tolerance for every d_Z , witnessing that the independent-mechanism inductive bias is compatible with (or actively helps) recovery; (c) for each sparsity the smallest N reaching averaged Amari ≤ 0.03 grows only polynomially in d_Z (fitted log-log slope $b \leq 3.0$); (d) at every (d_Z, s) the Amari(N) curve is nonincreasing modulo a 0.01 wobble tolerance.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: sparse_ica_learnability_summary.json: gates 4/4 true; sparse_ica_learnability_summary.json:status=pass

Source: experiments/sparse_ica_learnability/preregistration.json

squaring_separation

Hypothesis: For the monomial $z_n = x^{(2^n)}$ over $Mul = \{x, x\}$ and $Sq = \{x, x, sq\}$ with $sq(y)=yy$ definable, the four seams separate exactly: (US-1) both languages realise every positive degree; (US-2) every Mul -tree of degree 2^n has size $2^{n+1}-1$ while a Sq -tower has size $n+1$; (US-3) sharing-circuit size is n in both bases; (US-4) under the Gibbs prior $\pi(t)4^{-|t|}$, language-normalised fiber mass satisfies $\log_2(P_{Sq}/P_{Mul}) \geq 2(2^{n+1}-n-2)+\log_2(Z_{Mul}/Z_{Sq})-\log_2(C_{2^n-1})$, and at $n=4$ the Sq fiber occupies 27 size shells against one Catalan shell of 9694845 Mul -trees.

Decision rule: Accept the four-seam separation on this toy if and only if all six noncompensatory gates pass. Do not promote US-4 (EML gradient recovery).

Claim boundary: No explicit boundary field

Recorded signals: `squaring_separation_summary.json`: gates 6/6 true; `squaring_separation_summary.json`:status=pass

Source: `experiments/squaring_separation/preregistration.json`

structure_compiler

Hypothesis: A single substrate-independent dynamical structure (accumulation, phase transition, hysteresis) compiles into music, a visual field, text, and spatial navigation such that each medium's readback recovers the identical abstract trajectory, establishing verified cross-substrate structural identity rather than mood matching.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: `structure_compiler_summary.json`: gates 4/4 true; `structure_compiler_summary.json`:status=pass

Source: `experiments/structure_compiler/preregistration.json`

symbolic_causation

Hypothesis: Treating a symbolic model as an operation on a system's future-trajectory distribution, signal (KL), control (goal gain), knowledge (predictive accuracy), and agency (control plus calibrated self-attribution plus transfer) dissociate: no single scalar identifies agency, a false-credit intervention improves the outcome with zero true do-effect and large calibration error, and a brittle controller controls without transferring.

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: `symbolic_causation_summary.json`: gates 6/6 true; `symbolic_causation_summary.json`:status=pass

Source: `experiments/symbolic_causation/preregistration.json`

theory_atlas_pair

Hypothesis: Theorems TA-1 and TA-2 (paper papers/theory_atlas/paper.md) predict, on the 4-bit Boolean world with three context subsets $U_1=\{x_0=0\}$, $U_2=\{x_1=0\}$, $U_3=\{x_2=0\}$, chart maps $M_1(x)=g(x)$, $M_2(x)=(g(x)+1) \bmod 4$, $M_3(x)=(g(x)+2) \bmod 4$ for $g(x)=(2*x_2+x_3) \bmod 4$, and target label space $T=\mathbb{Z}/4$, that (i) the good transitions $T_{12}=+1$, $T_{23}=+1$, $T_{13}=+2$ satisfy the cocycle $T_{23} T_{12} = T_{13}$ on the single triple (1,2,3) and glue to the global theory $M=g$; (ii) the bad transitions $T_{12}=+1$, $T_{23}=+1$, $T_{13}=+3$ violate the cocycle with rank-4 discrepancy (shift-by-3 on $\mathbb{Z}/4$) and do not glue; and (iii) the bad-family failure has the missing-latent signature (rank ≥ 1 with all three T_{ij} non-identity), distinct from the phase-transition signature (rank ≥ 1 with exactly one T_{ij} non-identity) exhibited by a reference family ($T_{12}=+1$, $T_{23}=id$, $T_{13}=id$).

Decision rule: Use registered gates

Claim boundary: No explicit boundary field

Recorded signals: `theory_atlas_pair_summary.json`: gates 4/4 true; `theory_atlas_pair_summary.json`:status=pass

Source: `experiments/theory_atlas_pair/preregistration.json`

Appendix D - Exhaustive Lean declaration inventory

The source scan found 513 theorem, lemma, or proposition declarations across three Lean packages. It found no executable project-local axiom, sorry, admit, opaque declaration, or native_decide. A receipt naming a file does not automatically verify every declaration in that file; receipts enumerate headline subsets.

Source	Kind	Declaration	Lane / receipt signal
formal/relative-identifiability/RelativeIdentifiability.lean:39	theorem	indistinguishable_refl	relative: Lean-proved, no SafeVerify receipt
formal/relative-identifiability/RelativeIdentifiability.lean:46	theorem	indistinguishable_symm	relative: Lean-proved, no SafeVerify receipt
formal/relative-identifiability/RelativeIdentifiability.lean:54	theorem	indistinguishable_trans	relative: Lean-proved, no SafeVerify receipt
formal/relative-identifiability/RelativeIdentifiability.lean:116	theorem	factorsThrough_iff_fiberConstant	relative: Lean-proved, no SafeVerify receipt
formal/relative-identifiability/RelativeIdentifiability.lean:137	theorem	obstruction_iff_not_factors	relative: Lean-proved, no SafeVerify receipt
formal/relative-identifiability/RelativeIdentifiability.lean:167	theorem	indistinguishable_of_richer	relative: Lean-proved, no SafeVerify receipt
formal/relative-identifiability/RelativeIdentifiability.lean:189	theorem	forgetExperiments_surjective	relative: Lean-proved, no SafeVerify receipt
formal/relative-identifiability/RelativeIdentifiability.lean:199	theorem	indistinguishable_empty	relative: Lean-proved, no SafeVerify receipt
formal/relative-identifiability/RelativeIdentifiability.lean:206	theorem	constant_target_factors	relative: Lean-proved, no SafeVerify receipt
formal/structural-intelligence/StructuralIntelligence/AbstractionFrontier.lean:50	theorem	pareto_set_is_antichain	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/AbstractionFrontier.lean:77	theorem	pareto_contains_css_when_zero_sufficiency	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:84	theorem	kirchhoffPrediction_eq_sum_b	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:115	theorem	compose_id_left	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:124	theorem	compose_id_right	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:133	theorem	inverse_left	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:143	theorem	compose_noncomm	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:149	theorem	kirchhoff_flat_matches	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:153	theorem	kirchhoff_curved_matches	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:157	theorem	affine_A_escapes	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:161	theorem	affine_B_escapes	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:165	theorem	affine_C_escapes	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Aff13.lean:171	theorem	affine_escapes_kirchhoff	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/AlignmentGovernance.lean:48	theorem	viability_inherited_by_superset	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/AlignmentGovernance.lean:59	theorem	viability_valid_ext	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Antecedents.lean:43	theorem	intersection_refines_each	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Antecedents.lean:67	theorem	intersection_is_common_sufficient	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Antecedents.lean:87	theorem	intersection_is_coarsest_over_family	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/AutocatalyticAirtwork.lean:59	theorem	bayes_equals_boltzmann_with_reward_and_likelihood	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/AutocatalyticAirtwork.lean:66	theorem	bayesPosterior_eq_boltzmannUpdate	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Basic.lean:18	theorem	sumFin_zero	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Basic.lean:20	theorem	sumFin_succ	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Basic.lean:24	theorem	anyFin_zero	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Basic.lean:26	theorem	anyFin_succ	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Basic.lean:31	theorem	sumFin_zero_fn	core: Lean-proved; no file receipt located

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/Basic.lean:41	theorem	sumFin_const_one	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Basic.lean:49	theorem	one_le_sumFin_of_index	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Basic.lean:74	theorem	sumFin_add	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Basic.lean:83	theorem	sumFin_le_sumFin	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Basic.lean:92	theorem	anyFin_eq_true_iff	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CausalSemantics.lean:40	theorem	psiEquiv_refl	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CausalSemantics.lean:44	theorem	psiEquiv_symm	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CausalSemantics.lean:49	theorem	psiEquiv_trans	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CausalSemantics.lean:59	theorem	psi_equiv_preserves_under_context	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CausalSemantics.lean:87	theorem	messageQuotient_is_common_sufficient	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CausalSemantics.lean:95	theorem	messageQuotientMap_eq_iff	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CausalSemantics.lean:120	theorem	messageQuotient_is_coarsest	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CommonSuffScreen.lean:61	theorem	commonSuffScreen_refines	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CommonSuffScreen.lean:84	theorem	jointTaskQuotient_isCommonSuffScreen	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CommonSuffScreen.lean:96	theorem	jointTaskQuotient_eq_iff	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CommonSuffScreen.lean:122	theorem	commonSuffScreen_coarsest	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CommonSuffScreen.lean:138	theorem	commonSuffScreen_eq_jointTaskQuotient_iff	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:28	theorem	two_mul_pow	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:31	theorem	two_pow_add	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:51	theorem	degree_pos	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:56	theorem	size_succ_eq_two_mul_degree	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:64	theorem	size_of_pow2_degree	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:74	theorem	pow2Tree_degree	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:80	theorem	pow2Tree_size	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:109	theorem	sqTower_size	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:114	theorem	sqTower_degree	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:126	theorem	expand_degree	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:134	theorem	size_le_expand_size	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:162	theorem	getD_le_listMax	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:175	theorem	applyStep_le_two_max	core: Lean-proved; no file receipt located

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:194	theorem	listMax_append_singleton	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:202	theorem	listMax_append_le_two	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:220	theorem	degreesFrom_max_le	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:244	theorem	listMax_singleton_one	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:246	theorem	circuitMaxDegree_le_pow2	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:251	theorem	pow2_le_pow2_imp_le	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:254	theorem	circuit_pow2_needs_n_steps	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:262	theorem	repeatedSquaring_length	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:267	theorem	degreesFrom_length	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:278	theorem	degreesFrom_append	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:286	theorem	circuitDegrees_append	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:290	theorem	repeatedSquaring_reaches	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:337	theorem	tree_size_separation	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:346	theorem	circuit_size_of_pow2	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:355	theorem	conservative_extension	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Compiler/SquaringSeparation.lean:360	theorem	squaring_separation	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CompilerTomography.lean:74	theorem	not_consistent_iff_refuted	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CompilerTomography.lean:107	theorem	identifiability_implies_unique_by_witness	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CompilerTomography.lean:134	theorem	identifiability_yields_refuting_data	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CompilerTomography.lean:168	theorem	identifiability_isolates_theta_star	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:61	theorem	bag_representing_all_five	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:80	theorem	choice_delta_bag	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:83	theorem	choice_bag_first	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:86	theorem	choice_bag_last	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:89	theorem	choice_bag_pair_eq	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:92	theorem	choice_bag_parity	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:98	theorem	choice_all_six	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:104	theorem	unweighted_strictly_beaten	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:119	theorem	boundary_base	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:177	theorem	bag_ext_representing_seven	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:195	theorem	boundary_ext	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:203	theorem	mirrored_dual_ext	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernChoice.lean:212	theorem	boundary_menu_relative	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:62	theorem	seqBag_length	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:63	theorem	seqMix_length	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:64	theorem	seqPair_length	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:88	theorem	bag_stays	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:92	theorem	bag_step_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:99	theorem	mix_before_step	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:103	theorem	mix_stays	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:107	theorem	mix_step_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:118	theorem	pair_before_step	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:122	theorem	pair_stays	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:126	theorem	pair_step_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:133	theorem	misspec_sum_gap	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:138	theorem	misspec_expected_gap	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ConcernEstimate.lean:149	theorem	conc_est_registered_steps	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/CouponCollector.lean:39	theorem	theorem5_union_bound	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CouponCollector.lean:52	theorem	theorem5_deterministic_lower_bound	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/CrossingUnique.lean:24	theorem	crossing_unique	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:52	theorem	over_invariance_nogo	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:68	theorem	symmetry_mismatch_nogo	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:80	theorem	invariant_orbits_factor	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:95	theorem	identity_always_factors	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:116	theorem	path_integrates	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:128	theorem	prefixSum_cons	core: Lean-proved; no file receipt located

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:146	theorem	prefixSum_length_eq_sumInt	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:156	theorem	cycle_integrates_iff_sum_zero	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:163	theorem	potentials_unique_up_to_translation	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:200	theorem	repair_splits_disagreement	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DeleteRepair.lean:233	theorem	repair_paths_disagree	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/DialZero.lean:56	theorem	levelCells_zero_distortion	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DialZero.lean:70	theorem	zero_distortion_cell_in_level	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DialZero.lean:127	theorem	no_coarser_on_witness	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:132	theorem	interval_wMinus1	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:133	theorem	interval_wMinus8	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:135	theorem	interval_ne	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:141	theorem	wMinus1_incomparable	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:142	theorem	wMinus8_incomparable	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:148	theorem	eq_bool2_nat4	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:222	theorem	inPosetB_eq_diamondB_wMinus1	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:232	theorem	inPosetB_eq_diamondB_wMinus8	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:243	theorem	poset_wMinus1	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:248	theorem	poset_wMinus8	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:253	theorem	posetPairs_wMinus1	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:256	theorem	posetPairs_wMinus8	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:258	theorem	poset_eq	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DiamondInterval.lean:266	theorem	poset_not_determine_interval	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/DtaN4.lean:176	theorem	dta_n4_representable_iff	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:104	theorem	emlFib_counts	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:108	theorem	catalan_values	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:114	theorem	emlFib_total	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:153	theorem	emlVar_counts	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:158	theorem	emlVar_formula	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:164	theorem	emlVar_total	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:179	theorem	pair_same_size_distinct	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:197	theorem	left_denotes	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:206	theorem	right_denotes	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlCatalan.lean:234	theorem	eml_pair_diff	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/EmlZeroIdentity.lean:77	theorem	ln_one	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/EmlZeroIdentity.lean:83	theorem	eml_right_one	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/EmlZeroIdentity.lean:90	theorem	eml_eml_right_one	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/EmlZeroIdentity.lean:99	theorem	middle_pos	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/EmlZeroIdentity.lean:104	theorem	exp_ln_middle	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/EmlZeroIdentity.lean:109	theorem	eml_zero_identity	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/EmlZeroIdentity.lean:118	theorem	eml_zero_identity_one	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/EmlZeroIdentity.lean:123	theorem	eml_zero_identity_x	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:85	theorem	base7_count	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:87	theorem	ext7_count	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:89	theorem	base5_count	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:91	theorem	ext5_count	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:94	theorem	base_subset_ext7	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:97	theorem	base_subset_ext5	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:103	theorem	sq_min_base	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:105	theorem	sq_min_ext	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:107	theorem	sq_mass_base	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:109	theorem	sq_mass_ext	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:114	theorem	cube_min_base	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:116	theorem	cube_min_ext	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:118	theorem	cube_mass_base	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:120	theorem	cube_mass_ext	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:128	theorem	generator_border_sq	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:133	theorem	generator_border_cube	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/GeneratorBorder.lean:139	theorem	min_size_not_shared_function	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IcaSignedPerm.lean:65	theorem	id_signed	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IcaSignedPerm.lean:66	theorem	swap_signed	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IcaSignedPerm.lean:67	theorem	sign0_signed	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IcaSignedPerm.lean:68	theorem	swapSign_signed	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IcaSignedPerm.lean:69	theorem	dense_not_signed	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IcaSignedPerm.lean:71	theorem	id_acts_as_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IcaSignedPerm.lean:73	theorem	swap_swaps	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IcaSignedPerm.lean:75	theorem	sign0_flips_first	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IcaSignedPerm.lean:80	theorem	ica_class_fin2	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IdentImpossibility.lean:36	theorem	record_eq_of_pointwise	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IdentImpossibility.lean:51	theorem	no_transcript_map_hits_both	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IdentImpossibility.lean:69	theorem	hitCount_eq_zero_of_none	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IdentImpossibility.lean:85	theorem	hits_le_one	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IdentImpossibility.lean:139	theorem	registered_records_agree	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IdentImpossibility.lean:143	theorem	registered_separator_splits	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IdentImpossibility.lean:147	theorem	registered_no_passive_map	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/IdentImpossibility.lean:154	theorem	registered_hits_le_one	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:196	theorem	represents_qld	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:291	theorem	bag_has_nontrivial_symmetry	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:297	theorem	last_bit_has_nontrivial_symmetry	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:303	theorem	parity_has_nontrivial_symmetry	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:311	theorem	pair_eq_has_nontrivial_symmetry	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:321	theorem	allWorlds_length	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:323	theorem	bag_q_id_signature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:327	theorem	last_bit_q_id_signature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:331	theorem	parity_q_id_signature	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:335	theorem	pair_eq_q_id_signature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:339	theorem	bag_q_id_row_signature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:343	theorem	last_bit_q_id_row_signature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:347	theorem	parity_q_id_row_signature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:351	theorem	pair_eq_q_id_row_signature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:356	theorem	collision_rows_share_signature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:364	theorem	bag_q_id_gold	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:367	theorem	last_bit_q_id_gold	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:370	theorem	parity_q_id_gold	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:373	theorem	pair_eq_q_id_gold	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:376	theorem	bag_q_id_row_gold	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:379	theorem	last_bit_q_id_row_gold	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:382	theorem	parity_q_id_row_gold	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:385	theorem	pair_eq_q_id_row_gold	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:388	theorem	gold_quotient_ne_noop	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:392	theorem	kappaCheap_collisionSignature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaCheap.lean:400	theorem	kappa_cheap_not_function	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaRelabel.lean:240	theorem	yFirstBit_swap03	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaRelabel.lean:244	theorem	yLastBit_swap03	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaRelabel.lean:252	theorem	kappa_firstBit_qPerm	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaRelabel.lean:256	theorem	kappa_lastBit_qPerm	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaRelabel.lean:260	theorem	kappa_firstBit_qId	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaRelabel.lean:264	theorem	kappa_lastBit_qId	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaRelabel.lean:270	theorem	kappa_relabel_converse	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaRelabel.lean:278	theorem	kappa_relabel_natural	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:60	theorem	allWorlds_length	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:81	theorem	nameRank_is_lex_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:195	theorem	represents_of_isCommonSuffScreen	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:257	theorem	kirchhoffFlat_matches	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:260	theorem	affineA_mismatch	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:263	theorem	affineC_holonomy	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:266	theorem	affineC_kirchhoff	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:269	theorem	affineC_mismatch	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:356	theorem	suite_length	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:363	theorem	fiberCount_q_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:364	theorem	fiberCount_q_perm	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:365	theorem	fiberCount_q_rot	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:366	theorem	fiberCount_q_stab0	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:367	theorem	fiberCount_q_stab_last	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaScreen.lean:374	theorem	kappa_screen_hits_suite	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:142	theorem	preserves_yBag_represents	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:152	theorem	yBag_qId	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:155	theorem	yBag_qRot	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:160	theorem	yBag_qPerm	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:165	theorem	yBag_qStab0	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:170	theorem	yBag_qStabLast	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:174	theorem	qId_represents	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:177	theorem	qRot_represents	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:180	theorem	qPerm_represents	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:183	theorem	qStab0_represents	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:186	theorem	qStabLast_represents	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:190	theorem	each_menu_represents	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:212	theorem	representingScreens_eq_menu	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:216	theorem	bag_not_unique	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:226	theorem	uniqueExistential_fails	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KappaUnique.lean:237	theorem	uniqueness_also_fails_as_schedules	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/KleisliSection.I ean:42	theorem	sections_spec_indistinguishable	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KleisliSection.I ean:53	theorem	section_swap	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KleisliSection.I ean:113	theorem	four_sections_distinct	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/KleisliSection.I ean:123	theorem	four_sections_one_shadow	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MeaningVsCo mpany.lean:57	theorem	allContexts_complete	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MeaningVsCo mpany.lean:116	theorem	psiEqOn_iff_PsiEquiv	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MeaningVsCo mpany.lean:141	theorem	meaning_quotient_is_registered	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MeaningVsCo mpany.lean:156	theorem	company_quotient_is_parity	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MeaningVsCo mpany.lean:165	theorem	neither_partition_refines_the_other	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MeaningVsCo mpany.lean:173	theorem	company_does_not_refine_meaning	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MeaningVsCo mpany.lean:184	theorem	meaning_does_not_refine_company	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MenuBlind.lea n:156	theorem	fiberCount_pair01	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MenuBlind.lea n:158	theorem	fiberCount_pair23	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MenuBlind.lea n:164	theorem	gold_flip_pair_eq	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MenuBlind.lea n:172	theorem	gold_flip_pair23	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MenuBlind.lea n:183	theorem	menu_blind_kappa_impossible	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MenuBlind.lea n:203	theorem	base_gold_consistent	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/MenuBlind.lea n:211	theorem	screen_exact_on_flip_rows	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:54	theorem	allCharts_complete	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:61	theorem	allLabels_complete	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:109	theorem	allEdges_complete	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:188	theorem	taxon_exhaustive	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:193	theorem	taxon_mutually_exclusive	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:206	theorem	taxonomy_trichotomy	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:222	theorem	numNonIdentityEdges_le	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:227	theorem	not_hasIdentityEdge_iff	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:236	theorem	classify_glue_iff	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTa xonomy.lean:243	theorem	classify_boundary_iff	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:252	theorem	classify_missing_latent_iff	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:261	theorem	classify_conditions	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:306	theorem	good_classified_glue	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:309	theorem	phase_boundary_classified_boundary	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:312	theorem	bad_classified_missing_latent	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:315	theorem	good_discrepancy_rank_zero	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:318	theorem	bad_discrepancy_rank_four	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:321	theorem	phase_boundary_discrepancy_rank_four	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:324	theorem	bad_all_edges_non_identity	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:327	theorem	phase_boundary_has_identity_edge	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/ObstructionTaxonomy.lean:332	theorem	ta2_taxonomy_classifies	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/Pigeonhole.lean:29	theorem	countAt_nil	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Pigeonhole.lean:32	theorem	countAt_cons	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Pigeonhole.lean:36	theorem	one_le_countAt_of_mem	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Pigeonhole.lean:50	theorem	sumFin_indicator_eq_one	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Pigeonhole.lean:107	theorem	sumFin_countAt_eq_length	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Pigeonhole.lean:134	theorem	length_ge_of_covers	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Pigeonhole.lean:147	theorem	length_foldr_cons	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Pigeonhole.lean:158	theorem	mem_foldr_cons	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Pigeonhole.lean:189	theorem	coupon_collector_pigeonhole	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Refinement.lean:57	theorem	refinement_refl	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Refinement.lean:63	theorem	refinement_transitive	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Refinement.lean:75	theorem	qRel_refl	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Refinement.lean:79	theorem	qRel_symm	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Refinement.lean:83	theorem	qRel_trans	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Refinement.lean:89	theorem	qRel_refines_of_factor	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Refinement.lean:118	theorem	refinement_preserves_screen	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/Refinement.lean:138	theorem	refinement_preserves_screen_qRel	core: Lean-proved; no file receipt located

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:115	theorem	row_scalar_broken_misses	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:118	theorem	row_scalar_lifted_captures	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:121	theorem	row_scalar_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:169	theorem	row_grid_broken_misses	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:172	theorem	row_grid_lifted_captures	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:175	theorem	row_grid_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:205	theorem	row_fiber_broken_misses	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:208	theorem	row_fiber_lifted_captures	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:211	theorem	row_fiber_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:234	theorem	row_path_broken_misses	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:237	theorem	row_path_lifted_captures	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:240	theorem	row_path_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:264	theorem	row_proj_broken_misses	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:267	theorem	row_proj_lifted_captures	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:270	theorem	row_proj_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:300	theorem	row_ens_broken_misses	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:303	theorem	row_ens_lifted_captures	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:306	theorem	row_ens_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:353	theorem	row_iface_broken_misses	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:356	theorem	row_iface_lifted_captures	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:359	theorem	row_iface_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:388	theorem	row_gauge_broken_misses	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:391	theorem	row_gauge_lifted_captures	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:394	theorem	row_gauge_minimal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:436	theorem	rr1_row_scalar	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:441	theorem	rr1_row_grid	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:446	theorem	rr1_row_fiber	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.Jean:451	theorem	rr1_row_path	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/RepairTable.lean:456	theorem	rr1_row_proj	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.lean:461	theorem	rr1_row_ens	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.lean:466	theorem	rr1_row_iface	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.lean:471	theorem	rr1_row_gauge	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepairTable.lean:479	theorem	rr1_table_well_defined	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/RepresentationRepair.lean:62	theorem	liftEnsures_of_repairs_preserves	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/RepresentationRepair.lean:85	theorem	independent_lifts_compose_ensures	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/RepresentationRepair.lean:118	theorem	independent_lifts_compose	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:52	theorem	rearrange	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:72	theorem	tilt_pointwise	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:105	theorem	sumBy_le_sumBy	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:114	theorem	sumBy_mul_left	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:122	theorem	sumBy_mul_right	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:130	theorem	sumBy_add	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:139	theorem	cross_point	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:169	theorem	tilt_monotone	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:237	theorem	monitor_constant	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:277	theorem	witness_reward_strictly_rises	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:284	theorem	witness_value_strictly_falls	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SilentSubstitution.lean:296	theorem	silent_substitution_witness	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.lean:79	theorem	allWorlds_length	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.lean:81	theorem	mem_allWorlds	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.lean:119	theorem	swap23_mem	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.lean:121	theorem	swap23_ne_identity	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.lean:194	theorem	swap23_preserves_pair_eq	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.lean:284	theorem	q_perm_mixes_pair	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.lean:287	theorem	q_stab0_mixes_pair	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.lean:290	theorem	q_stab_last_mixes_pair	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:294	theorem	q_rot_mixes_pair	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:297	theorem	mixes_pair_eq_q_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:298	theorem	mixes_pair_eq_q_rot	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:299	theorem	mixes_pair_eq_q_perm	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:300	theorem	mixes_pair_eq_q_stab0	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:301	theorem	mixes_pair_eq_q_stab_last	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:303	theorem	nFibres_q_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:305	theorem	hasNontrivialSymmetry_pair_eq	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:308	theorem	pair_eq_q_id_signature	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:317	theorem	nat_blt_16_16	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:319	theorem	menu_any_unfold	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:324	theorem	menu_any_coarser_representing_pair_eq _q_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:337	theorem	unused_symmetry_not_privilege	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:350	theorem	goldOf_pair_eq_q_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:373	theorem	decide_pair_eq_q_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SurgeryMiss.I ean:379	theorem	surgery_miss_pair_eq	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SwapTyped.Ie an:103	theorem	over_typed_stab0	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SwapTyped.Ie an:104	theorem	over_typed_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SwapTyped.Ie an:105	theorem	over_crossed_perm	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SwapTyped.Ie an:106	theorem	under_typed_perm	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SwapTyped.Ie an:107	theorem	under_privilege_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SwapTyped.Ie an:108	theorem	fiber_perm_lt_id	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/SwapTyped.Ie an:112	theorem	swap_typed_wins	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/T4FiniteCI.lea n:42	theorem	css_implies_fiber_constant	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/T4FiniteCI.lea n:83	theorem	firstBit_is_css	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/T4FiniteCI.lea n:89	theorem	registered_fiber_constant	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/T4FiniteCI.lea n:98	theorem	secondBit_not_fiber_constant	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/TheoryAtlas.Ie an:76	theorem	cocycle_implies_gluing	core: Lean-proved; no file receipt located

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/TheoryAtlas.lean:97	theorem	injective_gluing_implies_cocycle	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/UnionBound.lean:26	theorem	indicator_anyFin_le_sumFin_indicator	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/UnionBound.lean:49	theorem	countP_anyFin_le_sumFin_countP	core: Lean-proved; no file receipt located
formal/structural-intelligence/StructuralIntelligence/WeakestAdequate.lean:60	theorem	both_adequate	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeakestAdequate.lean:65	theorem	union_not_adequate	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeakestAdequate.lean:70	theorem	both_maximal	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeakestAdequate.lean:80	theorem	no_largest_adequate	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeakestAdequate.lean:97	theorem	maximal_not_unique	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeakestAdequate.lean:112	theorem	greedy_repair_works	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeakestAdequate.lean:122	theorem	greedy_depends_on_order	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:66	theorem	shortcut_in_ge1	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:67	theorem	shortcut_not_ge2	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:68	theorem	invariant_in_ge1	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:69	theorem	invariant_in_ge2	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:71	theorem	card_ge1	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:72	theorem	card_ge2	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:74	theorem	shortcut_mass	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:75	theorem	invariant_mass	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:77	theorem	cert_shortcut_k1	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:78	theorem	cert_invariant_k1	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:79	theorem	cert_invariant_k2	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessMixture.lean:87	theorem	mixture_prior_mass_increases	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessP1.lean:88	theorem	shortcut_train_ok	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessP1.lean:89	theorem	invariant_train_ok	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessP1.lean:91	theorem	shortcut_weakness_one	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessP1.lean:92	theorem	invariant_weakness_two	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessP1.lean:94	theorem	shortcut_covers_one	core file named in receipt; headline subset only
formal/structural-intelligence/StructuralIntelligence/WeaknessP1.lean:95	theorem	invariant_covers_two	core file named in receipt; headline subset only

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence/StructuralIntelligence/WeaknessP1.lean:100	theorem	coverage_increases	core file named in receipt; headline subset only
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/AA1MonotoneCompetence.lean:64	theorem	aa1_log_mixture_ge_weighted_log	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/AA1MonotoneCompetence.lean:94	theorem	aa1_log_mixture_ge_weighted_log_sample	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/AA1MonotoneCompetence.lean:119	theorem	aa1_refinement_raises_lower_bound	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/AG1Survival.lean:56	theorem	ag1_survival_lower_bound	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/AG1Survival.lean:81	theorem	ag1_joint_survival	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/AG1Survival.lean:98	theorem	ag1_joint_survival_linear	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG1FisherMatrix.lean:98	theorem	hasDerivAt_expFamZ	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG1FisherMatrix.lean:125	theorem	cg1_logpartition_deriv_eq_meanT	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG1FisherMatrix.lean:146	theorem	expFamP_pos	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG1FisherMatrix.lean:170	theorem	cg1_score_identity_scalar	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG1FisherMatrix.lean:215	theorem	cg1_fisher_eq_variance	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG1FisherMatrix.lean:227	theorem	expFamScore_mean_zero	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG1FisherMatrix.lean:315	theorem	cg1_fisher_matrix_eq_covariance	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG2Holonomy.lean:85	theorem	cg2_holonomy_equals_signed_area	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG2Holonomy.lean:104	theorem	cg2_discrete_greens_grid	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG2Holonomy.lean:148	theorem	cg2_discrete_greens_symmetric	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG2Holonomy.lean:165	theorem	cg2_bottom_edge_riemann	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG2Holonomy.lean:179	theorem	cg2_top_edge_riemann	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CG2Holonomy.lean:199	theorem	cg2_boundary_riemann_equals_area	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CT2MonotoneReward.lean:56	theorem	chebyshev_weighted_correlation	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CT2MonotoneReward.lean:169	theorem	exp_beta_comonotone	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CT2MonotoneReward.lean:206	theorem	ct2_covariance_nonneg	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/CT2MonotoneReward.lean:231	theorem	ct2_boltzmann_raises_expected_reward	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Proposition3Adjunction.lean:65	theorem	coarsen_refine_eq	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Proposition3Adjunction.lean:129	theorem	R_C_unit	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Proposition3Adjunction.lean:141	theorem	C_R_counit	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Proposition3Adjunction.lean:161	theorem	proposition3_adjunction	mathlib: proved, not SafeVerify-verified

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/SICA_FiniteExistence.lean:89	theorem	sic_a_finite_discrete.	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/SICA_FiniteExistence.lean:165	theorem	sic_a_finite_discrete_coarsest	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/SICC_CoveringMeta.lean:91	theorem	sicc_covering_meta	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/SICC_CoveringMeta.lean:123	theorem	sicc_covering_poly	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:25	lemma	n_cast_ne_zero	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:29	lemma	binaryEntropy_eq_binEntropy	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:35	lemma	log_nsub_int	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:38	lemma	rdPhi_eq_qary	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:47	lemma	mul_log_mul_of_nonneg	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:62	lemma	jointEntropy_chain	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:93	lemma	mutualInfo_eq_source_minus_rev	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:102	lemma	entropy_le_log_card	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:158	lemma	entropy_split_error	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:288	lemma	uniform_nonneg	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:292	lemma	uniform_sum	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:298	lemma	marginal_nonneg	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:304	lemma	marginal_sum	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:320	lemma	posterior_nonneg	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:331	lemma	posterior_sum	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:342	lemma	expected_hamming_row	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:369	lemma	expected_hamming_eq	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:378	lemma	posterior_diag	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:397	lemma	weighted_error_eq_expected	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:415	lemma	condEntropyRev_eq_avg	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:504	lemma	fano_avg	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:581	lemma	qary_mono	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:596	theorem	Shannon1959_converse_uniform_hamming	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/ShannonFano.lean:624	theorem	R_D_uniform_hamming	mathlib: proved, not SafeVerify-verified

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem1MinimalSufficiency.lean:93	theorem	IsSufficient_iff_likelihood_ratio_factors	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem1MinimalSufficiency.lean:155	theorem	likelihoodRatioVector_sufficient	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem1MinimalSufficiency.lean:217	theorem	HalmosSavage_minimality_h_extension	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem1MinimalSufficiency.lean:255	theorem	exists_minimal_sufficient_finite_discrete	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:92	lemma	nsub_one_pos	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:96	lemma	nsub_one_ne_zero	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:99	lemma	n_pos	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:103	lemma	n_ne_zero	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:113	theorem	symChannel_stochastic_row	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:138	theorem	symChannel_stochastic_col	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:159	theorem	symChannel_marginal_uniform	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:176	theorem	symChannel_expected_hamming	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:224	theorem	symChannel_row_entropy	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:276	theorem	symChannel_mutualInfo_closed_form	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:323	lemma	binaryEntropy_zero	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:328	lemma	entropy_dirac	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:342	lemma	entropy_uniform	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:360	lemma	hammingDistortion_nonneg	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem2RateDistortion.lean:372	theorem	Shannon1959_converse_D_zero	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem5Rate.lean:50	theorem	theorem5_rate_bound	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem5Rate.lean:103	theorem	one_sub_le_exp_neg_pow	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/Theorem5Rate.lean:125	theorem	one_sub_inv_pow_le_exp_neg_div	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/WeaknessPACBayes.lean:59	lemma	mem_Hge_of_le	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/WeaknessPACBayes.lean:63	lemma	Hge_nonempty_of_le	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/WeaknessPACBayes.lean:67	lemma	mixtureMass_ge_component	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/WeaknessPACBayes.lean:98	theorem	weakness_kl_certificate	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/WeaknessPACBayes.lean:122	theorem	lsm_plug_certificate	mathlib: proved, not SafeVerify-verified
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/WeaknessPACBayes.lean:136	lemma	lsmExtra_nonneg	mathlib: proved, not SafeVerify-verified

Source	Kind	Declaration	Lane / receipt signal
formal/structural-intelligence-mathlib/StructuralIntelligenceMathlib/WeaknessPACBayes.lean:154	theorem	weakness_lsm_bound	mathlib: proved, not SafeVerify-verified

Appendix E - Audit limitations and reproduction contract

This atlas audits committed repository evidence at one Git commit. It does not independently rerun every historical model provider, reconstruct gitignored raw artifacts, validate every external citation, or prove that a paper's prose follows from its formal statement. Provider-backed results are snapshots. Exact finite results are only exact for their declared world and code.

```
Rebuild command: python3 scripts/build_formal_system_atlas_pdf.py
```

Primary verification commands used by repository policy include `python3 scripts/run_quality_checks.py` and the Lean lake builds in `.github/workflows/quality.yml`. SafeVerify status must be read from the dated receipts under `docs/lea` rather than inferred from a green lake build.

The generated appendices should change when registries, preregistrations, result status signals, or Lean declarations change. Such drift is expected and is why the PDF records its commit and date.